Why is climate change an economic problem?

ECON-4210
Economics of Climate Change
Class 4

Prof. Casey Wichman

Georgia Institute of Technology

Today's class

1. Questions?
2. Attendance
3. **Climate change as an economic problem**
4. For next class…

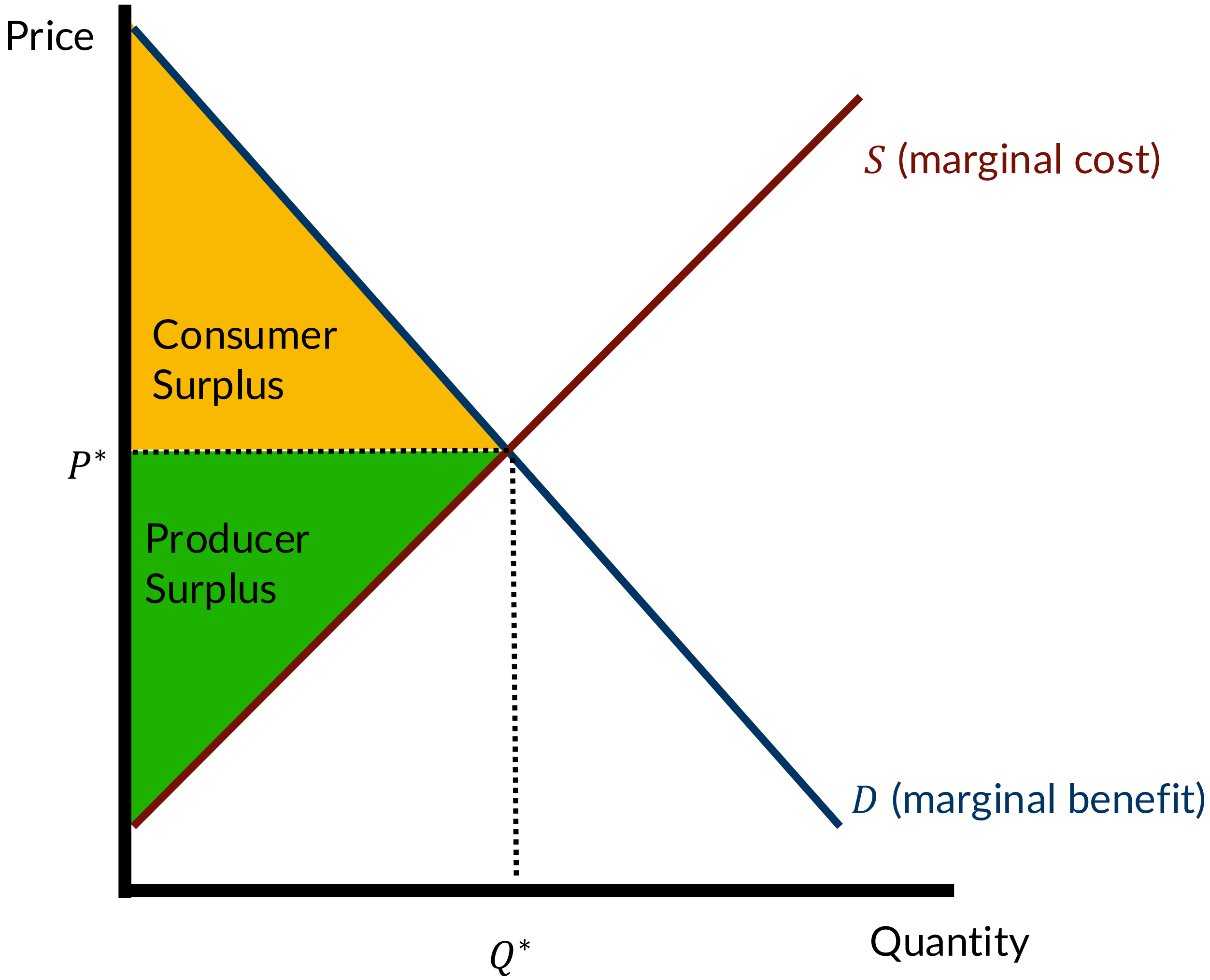
The goal of markets is to *efficiently* allocate scarce resources

Economic efficiency = maximizing net surplus

What does this mean?

What is economic efficiency?

Maximizing the size of the pie.



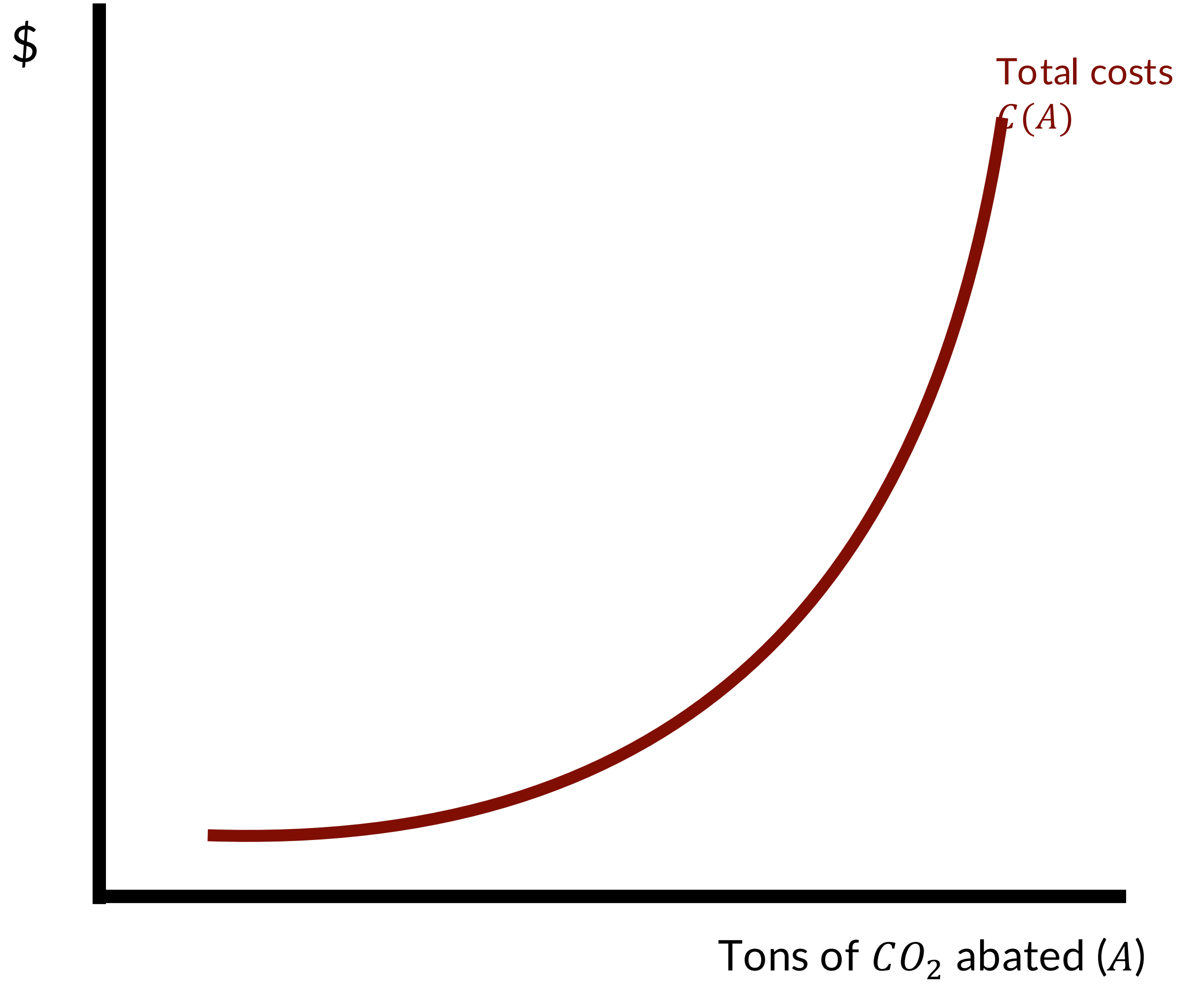
Refresher on supply & demand

- The supply curve represents **marginal costs** of producers.
- The demand curve represents the **marginal benefits** of consumers.
- The area <u>above</u> the supply curve but below the equilibrium price is **producer surplus**.
- The area <u>below</u> the demand curve but above the equilibrium price is **consumer surplus**.

- Equilibrium outcomes (Q^*, P^*) maximize economic surplus (CS + PS).

- There's actually a lot of economics behind these two curves.
- Let's build up intuition for why this is the case.

Costs of reducing GHG emissions



- CO_2 emissions are a "bad." They cause climate change.
- ***CO_2 abatement (A)*** is a "good." Abatement reduces CO_2 emissions that cause climate change.
- We can model CO_2 abatement like any other economic good.

- Why does the total cost curve of abatement increase at an increasing rate

- Is this realistic for CO_2?

Cost of reducing GHG emissions

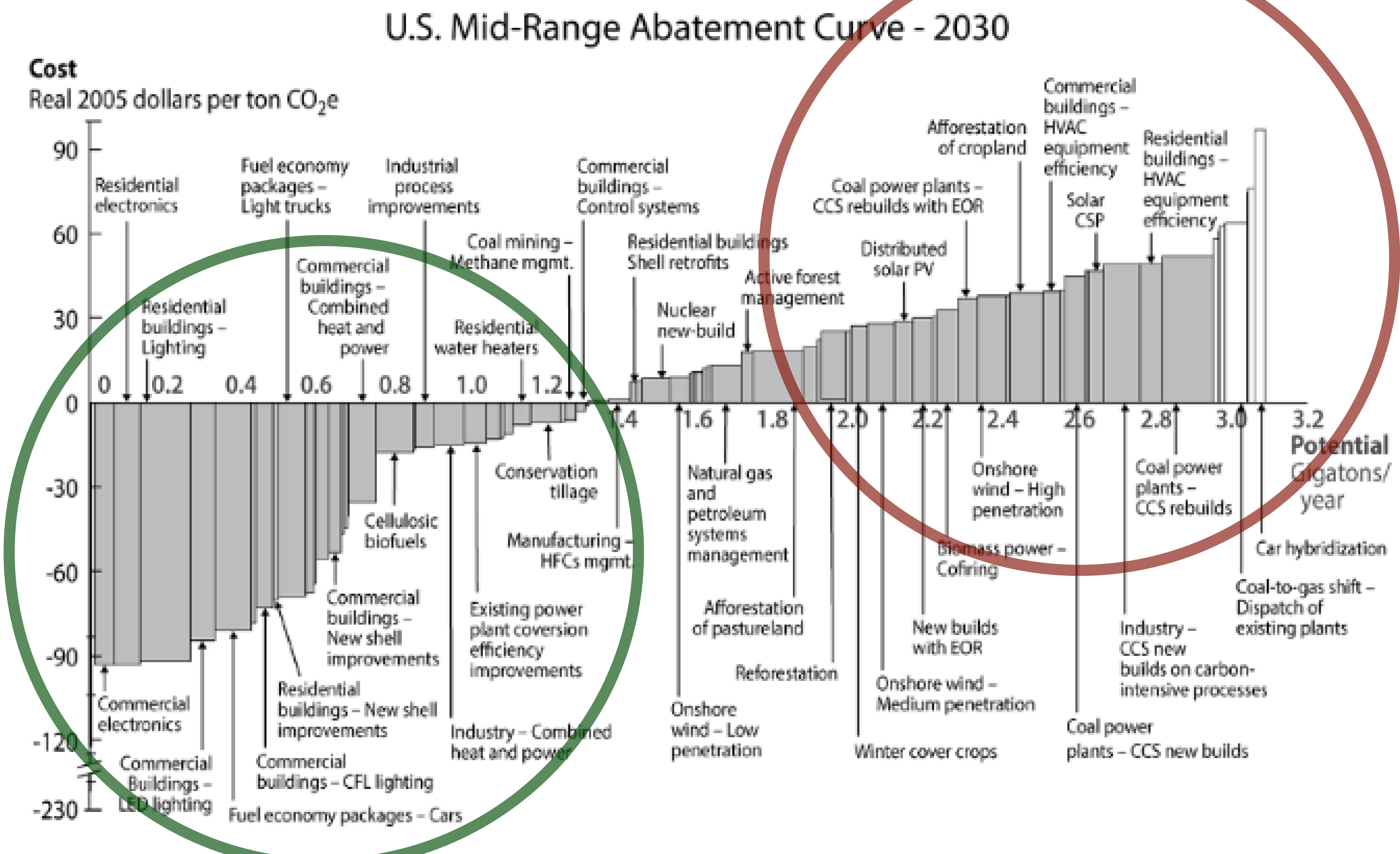
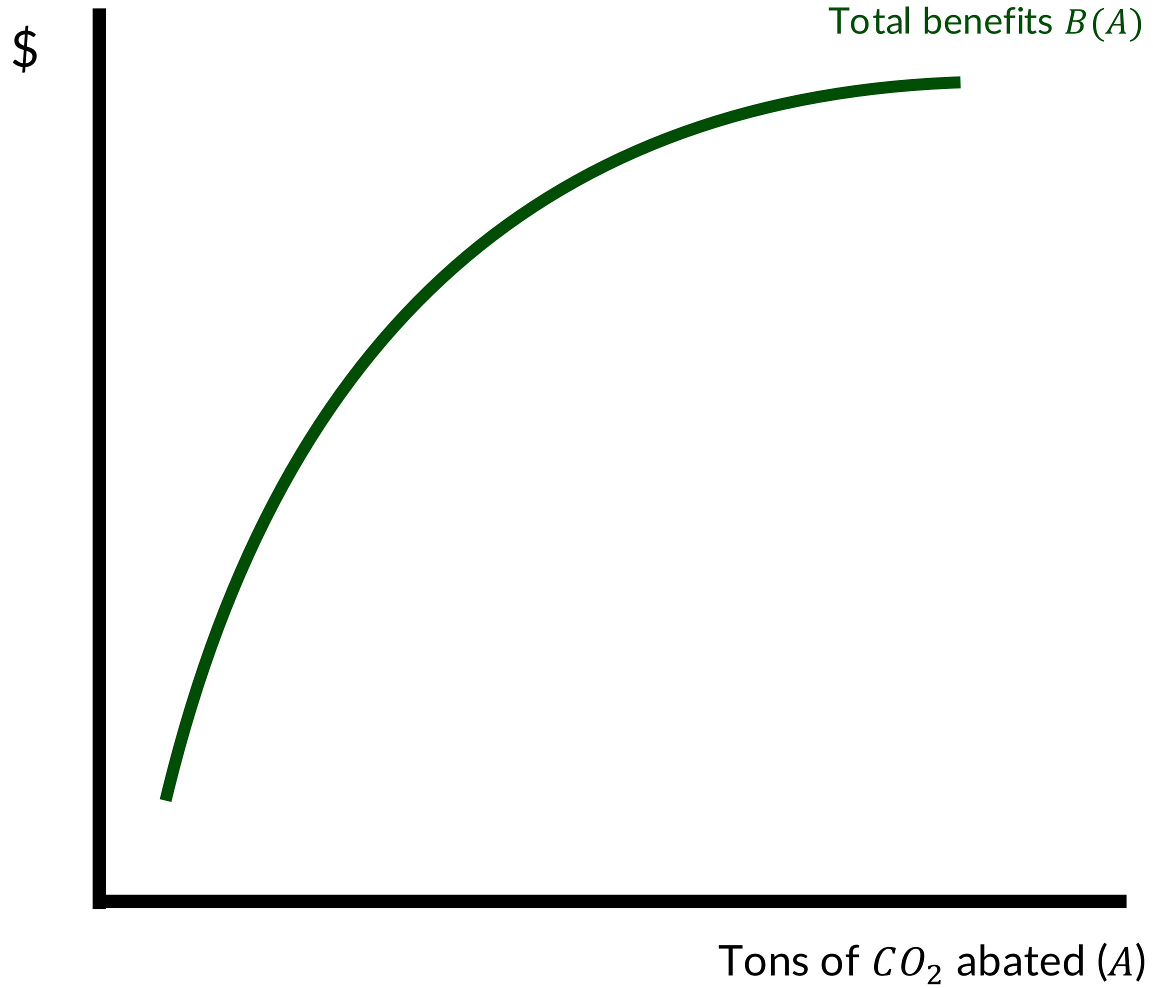


Figure 3.1 U.S. marginal abatement cost curve for greenhouse gases, developed by McKinsey & Company.

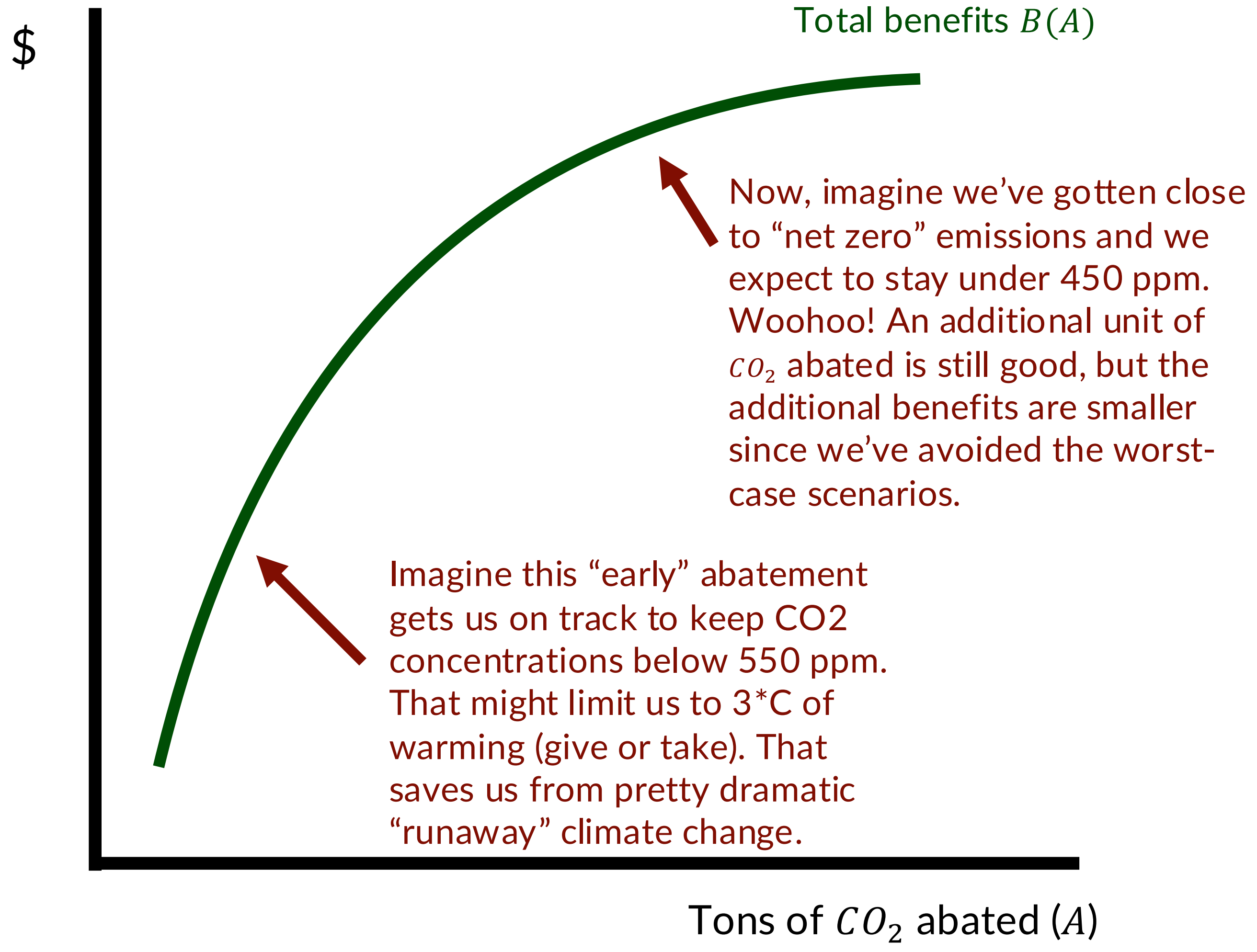
Source: Keohane & Olmstead, Ch. 3 (2016)

Benefits of reducing GHG emissions



- Abating (or reducing) GHG emissions provides benefits, namely less climate change.
- Think of GHG abatement on a range from "business as usual" emissions to "net zero" (no effective CO_2 emissions)
- **Why do benefits increase at a decreasing rate?**
- **Is this realistic for GHG abatement?**

Benefits of reducing GHG emissions



- Abating (or reducing) GHG emissions provides benefits, namely less climate change.
- Think of GHG abatement on a range from "business as usual" emissions to "net zero" (no effective CO_2 emissions)
- **Why do benefits increase at a decreasing rate?**
- **Is this realistic for GHG abatement?**

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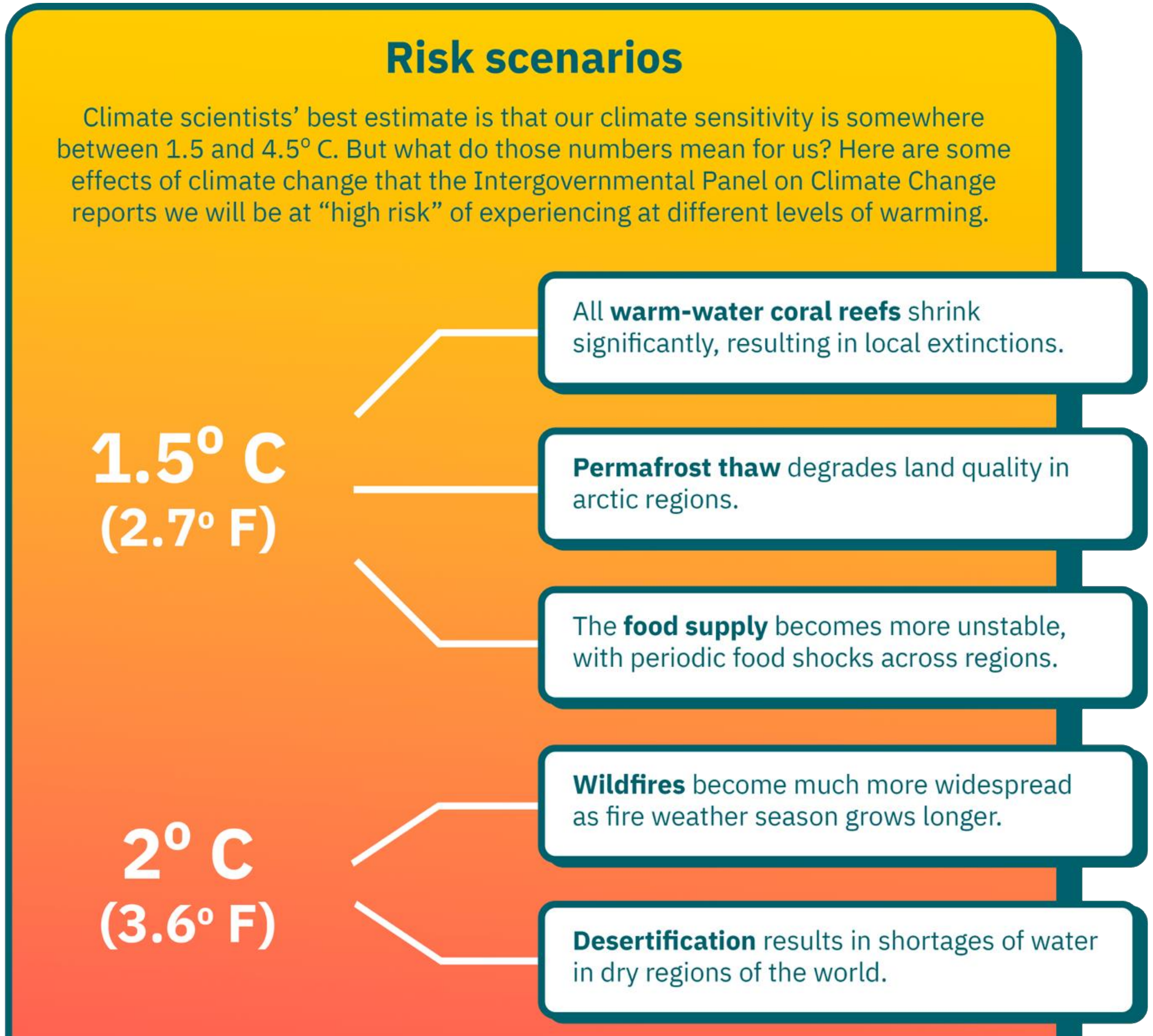
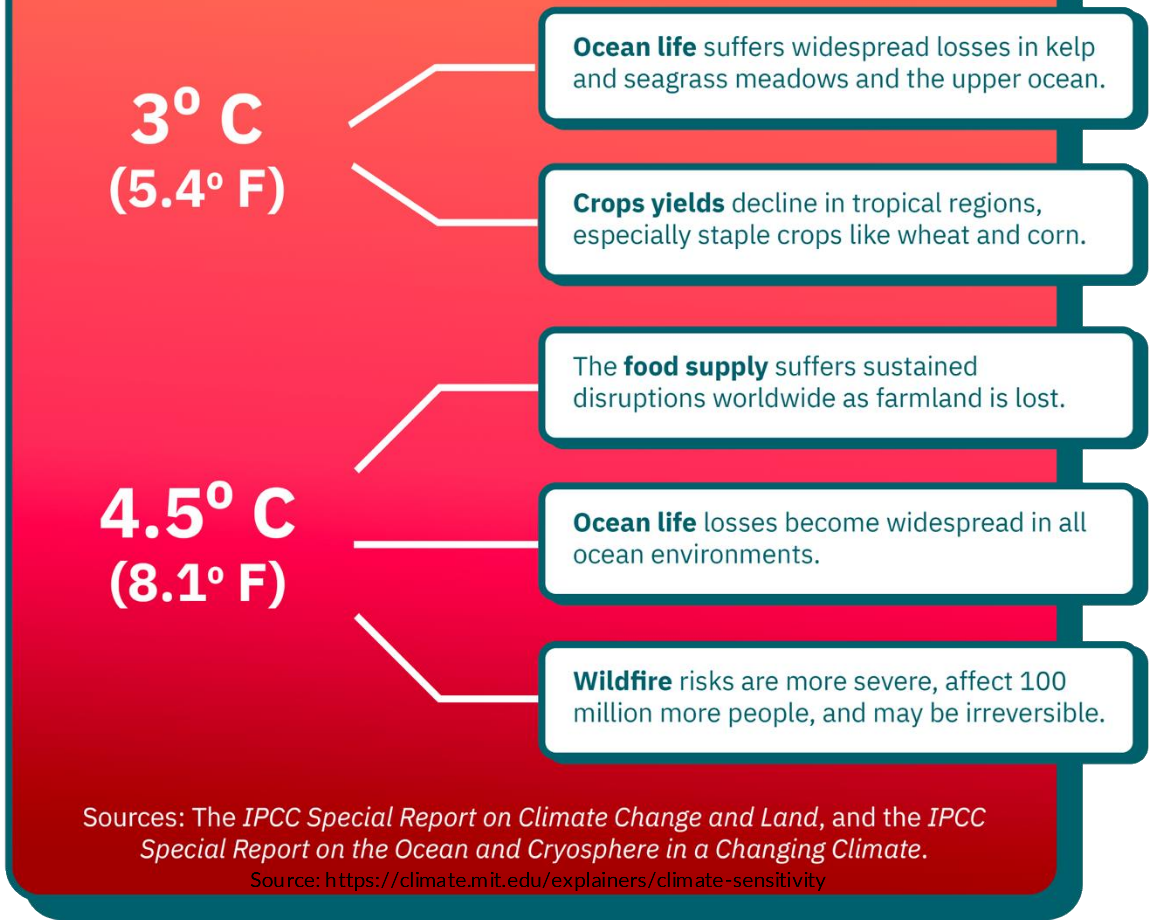
Think of this list (backwards) as emblematic of the benefits of GHG abatement





Source: https://climate.mit.edu/explainers/climate-sensitivity

- i.e., if we abate enough GHGs to stay under 3°C, we can avoid some these (really bad) consequences.

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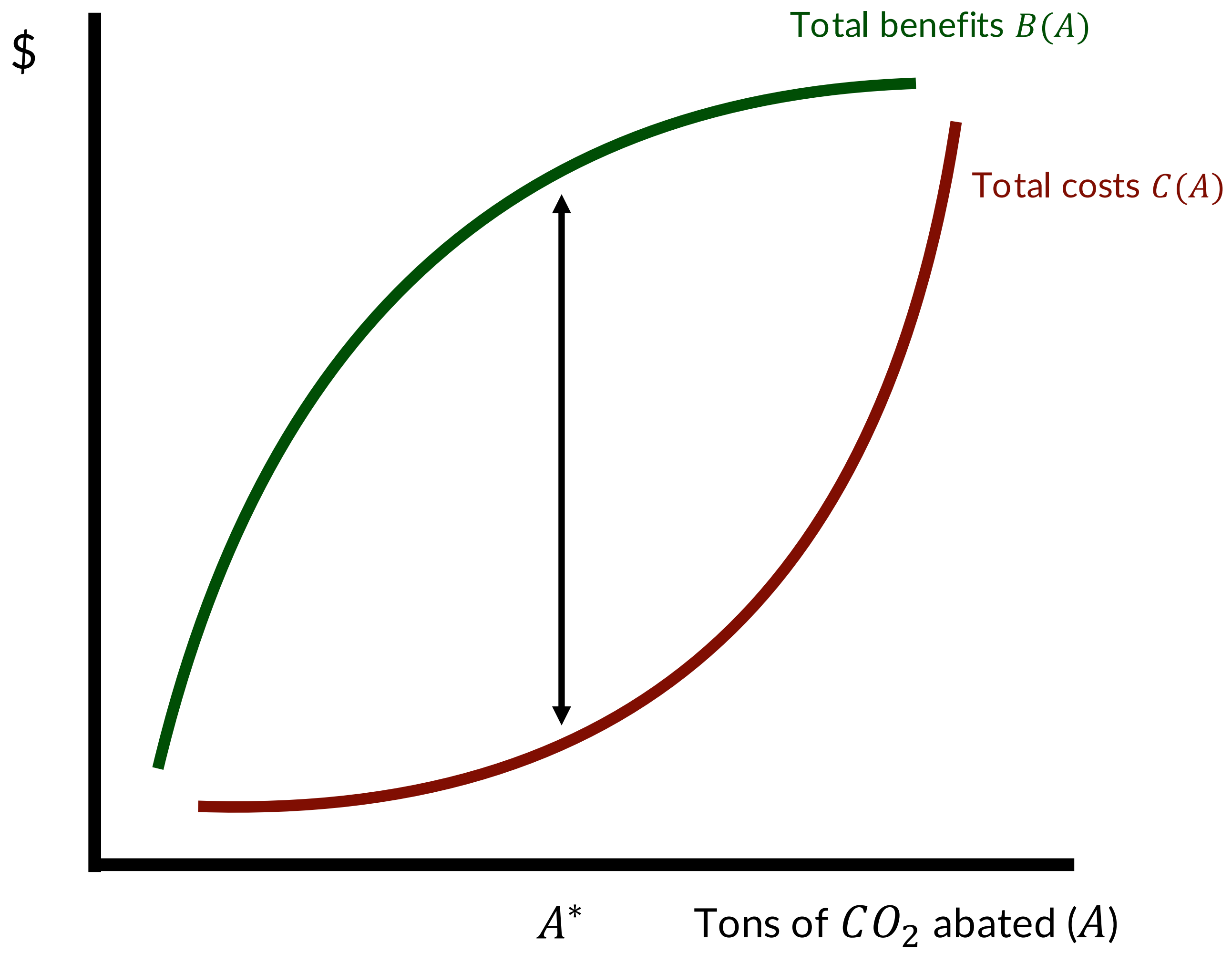
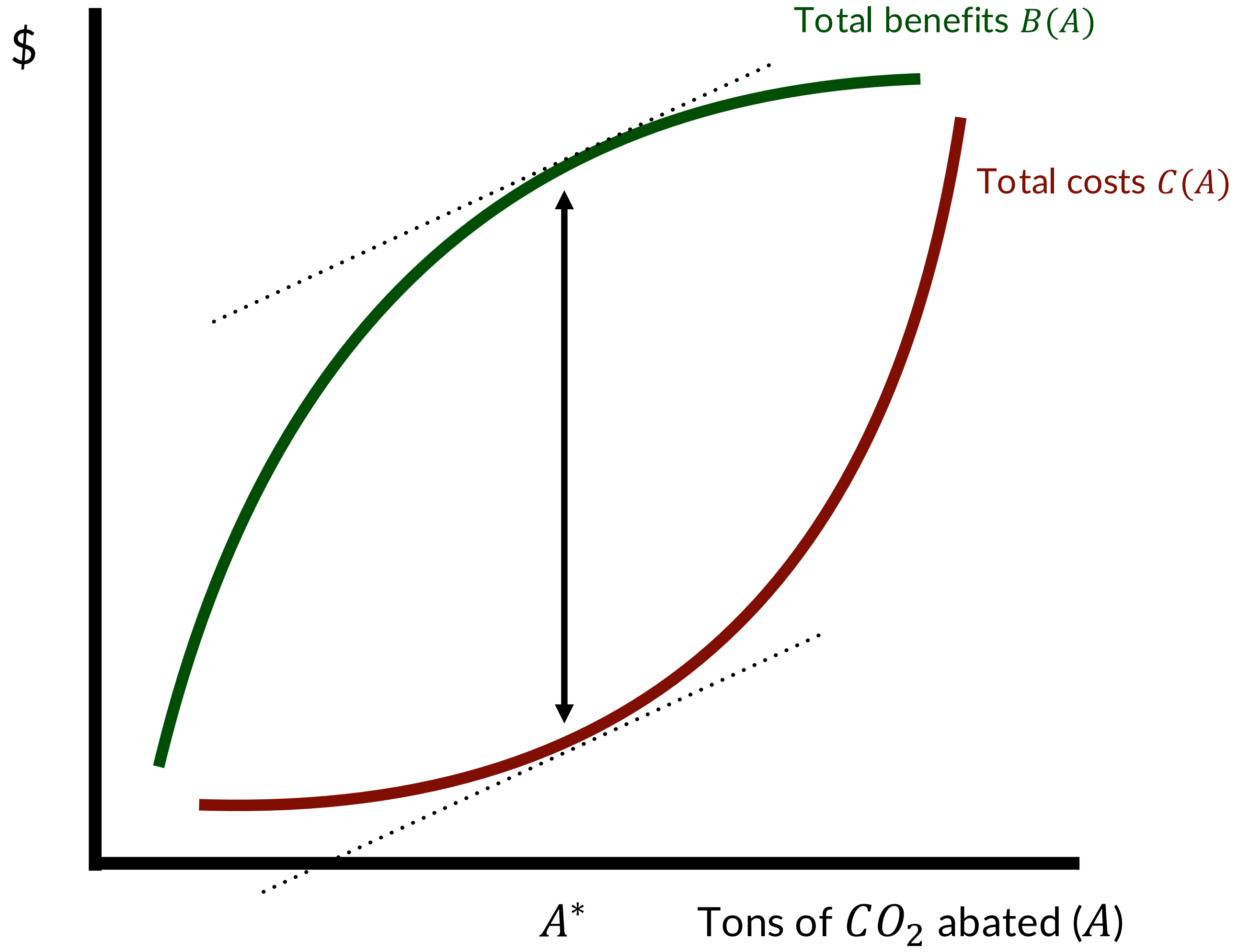
Maximizing net benefits



- We want to adopt a policy that maximize net benefits of reductions of CO_2 .That will be the efficient allocation of resources.
- So, we find the level of abatement (A^*) where $B(A) - C(A)$ is largest.
- A key insight here is that the optimal amount of abatement is typically not abating ALL of the pollution
- **Why is this the case?**
- For GHGs, it's possible we might actually want to abate all emissions (and then some).

So why do we usually work with supply/demand curves rather than total cost and total benefit curves?

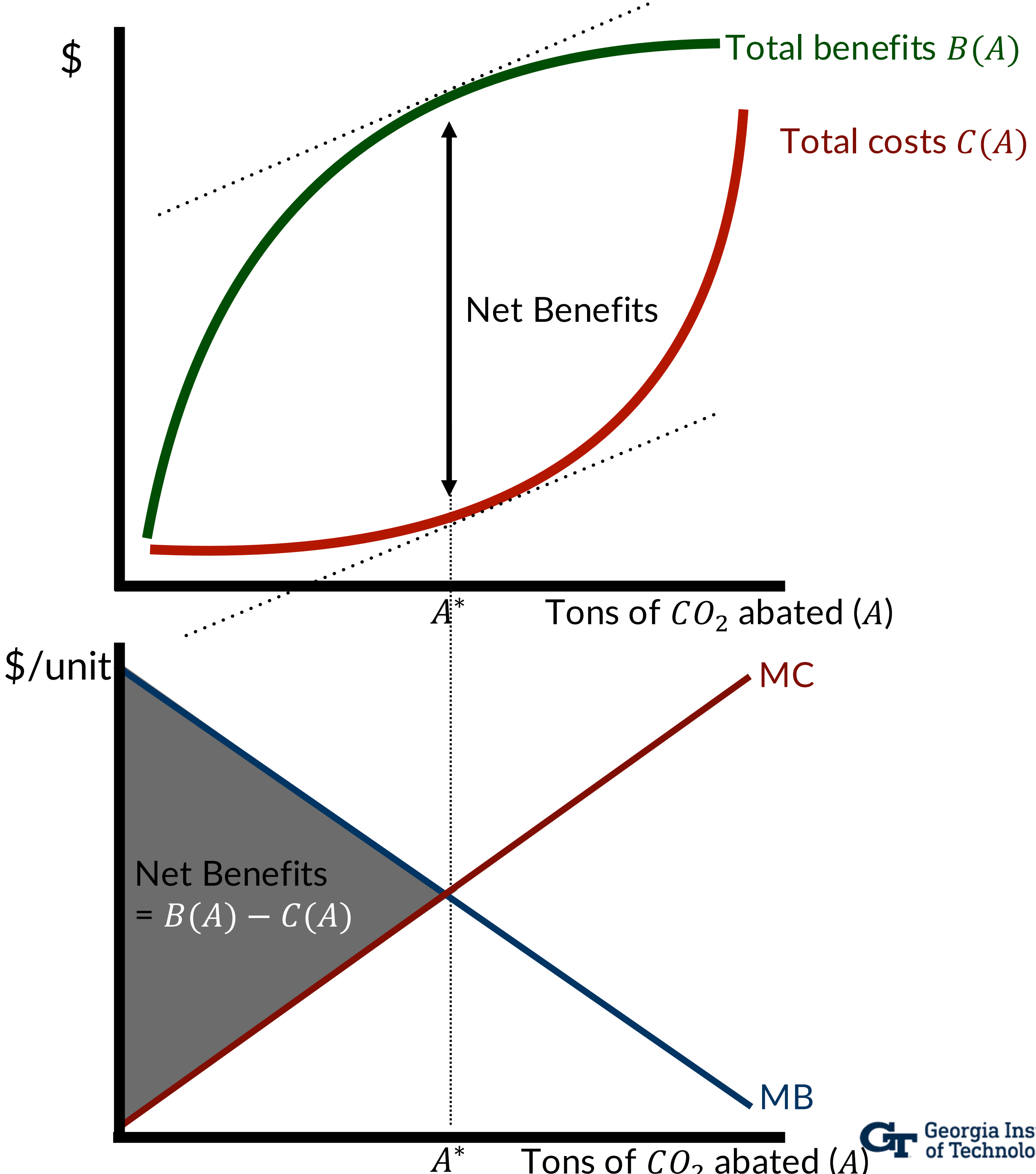


- In "ECON 101" we were taught to set *marginal* costs equal to *marginal* benefits.
  - Why?
- Because that maximizes total net benefits.
  - Why?

- The marginal cost curve is just the slope (or derivative) of the total cost curve.
- The marginal benefit curve is just the slope (or first derivative) of the total benefit curve.
  - $max_A B(A) - C(A)$
  - $\rightarrow B'(A^*) = C'(A^*)$
  - $\Longrightarrow MB(A^*) = MC(A^*)$

From net benefits to supply and demand

- **MB** is decreasing because $B(A)$ is increasing at a decreasing rate
    - This is referred to as "diminishing marginal utility"
- **MC** is increasing because $C(A)$ is increasing at an *increasing* rate.
    - This is referred to as "diminishing marginal returns"
- Why is that gray triangle equal to net benefits?



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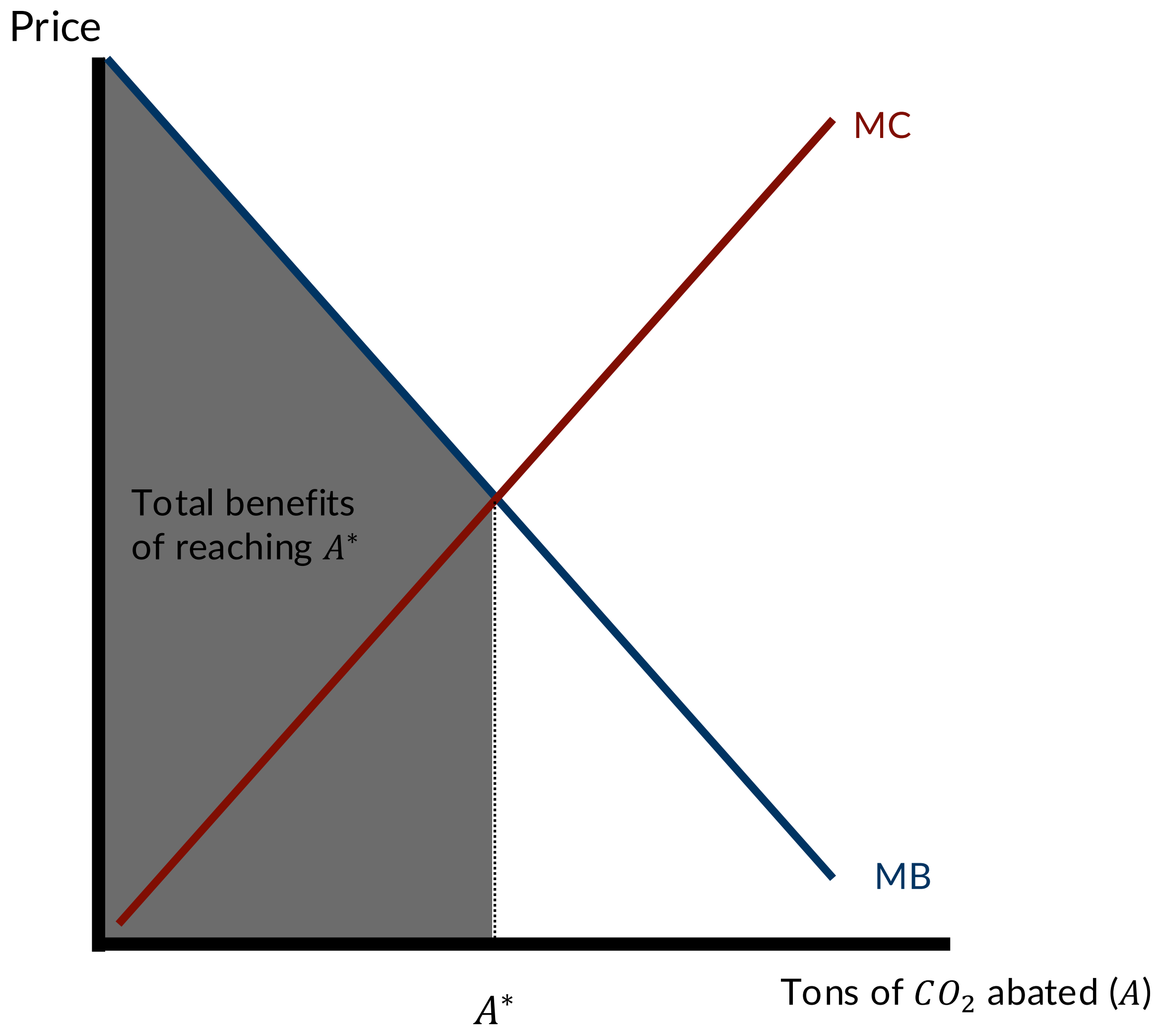
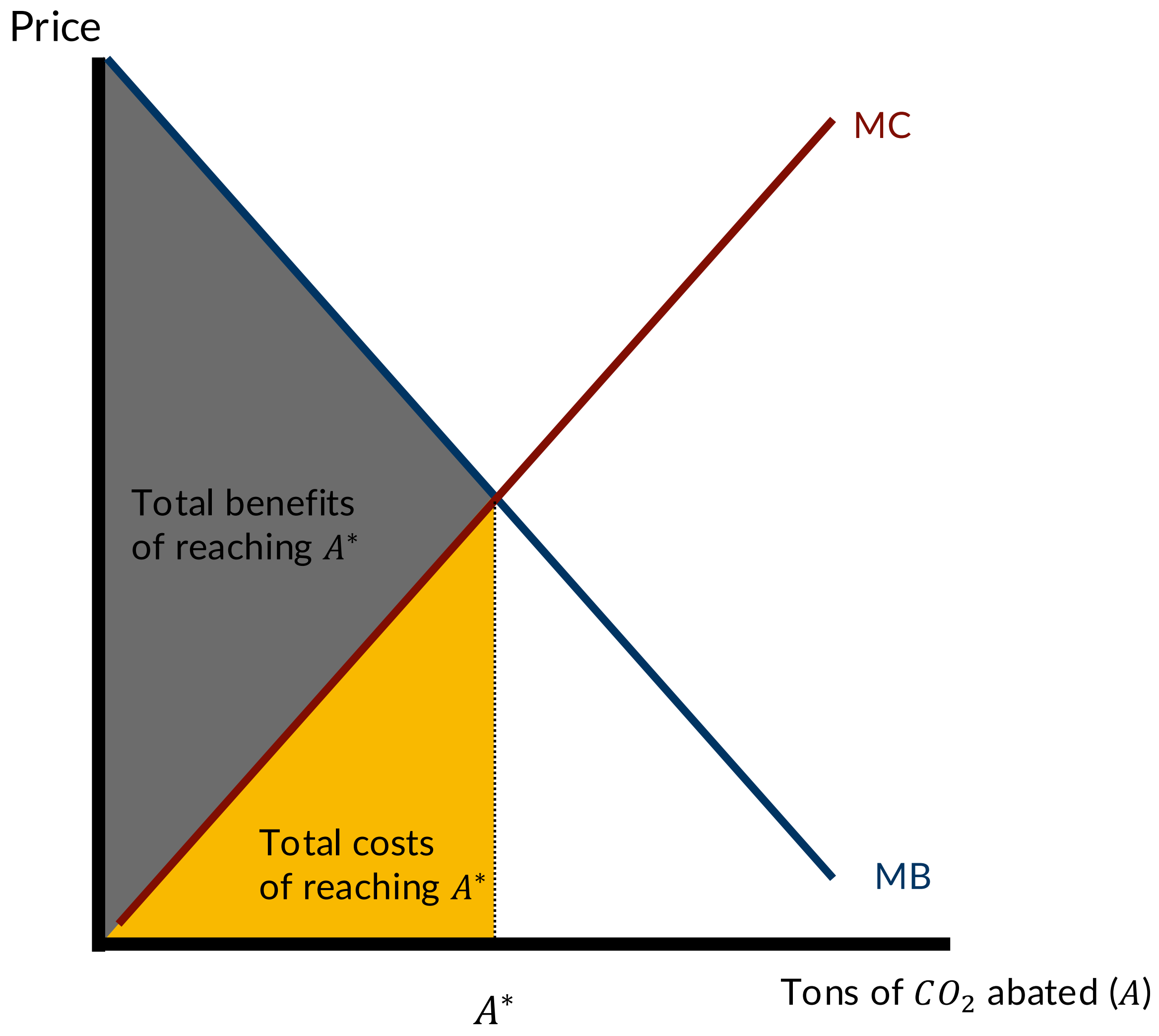
Net benefits (behind the scenes…)



- That gray area is **Total Benefits** from providing abatement level A*
  - We're just finding the area (i.e., integrating) underneath the marginal benefit curve to "get back to" total benefits.

Net benefits (behind the scenes…)



- That gray area is **Total Benefits** from providing abatement level A*
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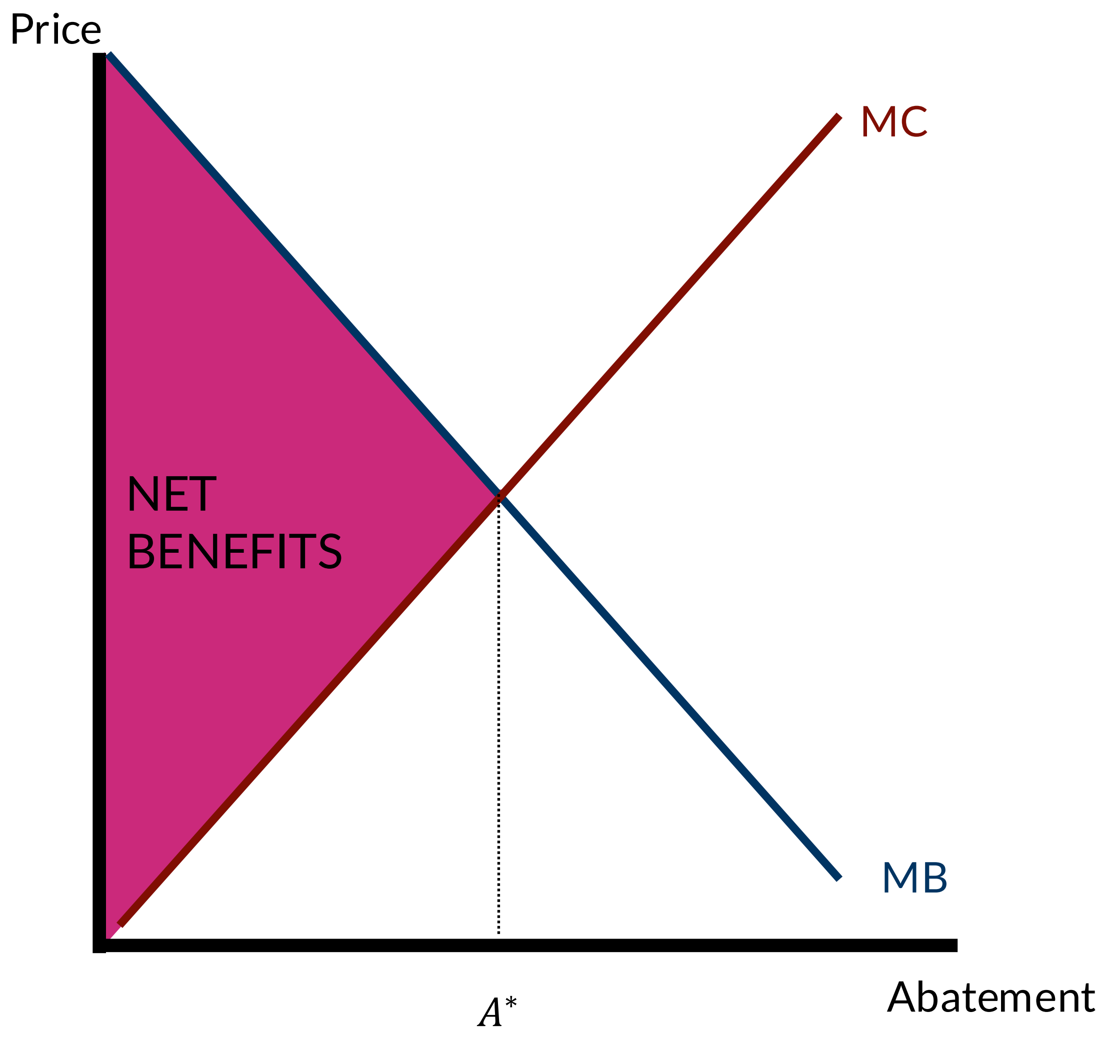
- The yellow area is **Total Costs** from providing abatement level A*
  - Again, this area is also an integral.

Net benefits (behind the scenes…)



- That gray area is **Total Benefits** from providing abatement level A*
  - We're just finding the area (i.e., integrating) underneath the marginal benefit curve to "get back to" total benefits.
- The yellow area is **Total Costs** from providing abatement level A*
  - Again, this area is also an integral.
- Net benefits is just subtracting the area representing **total costs** from the area representing **total benefits**.
- Note, you don't need calculus to understand any of this for this course! But it might help to know that there is very intuitive and straightforward math behind these figures…

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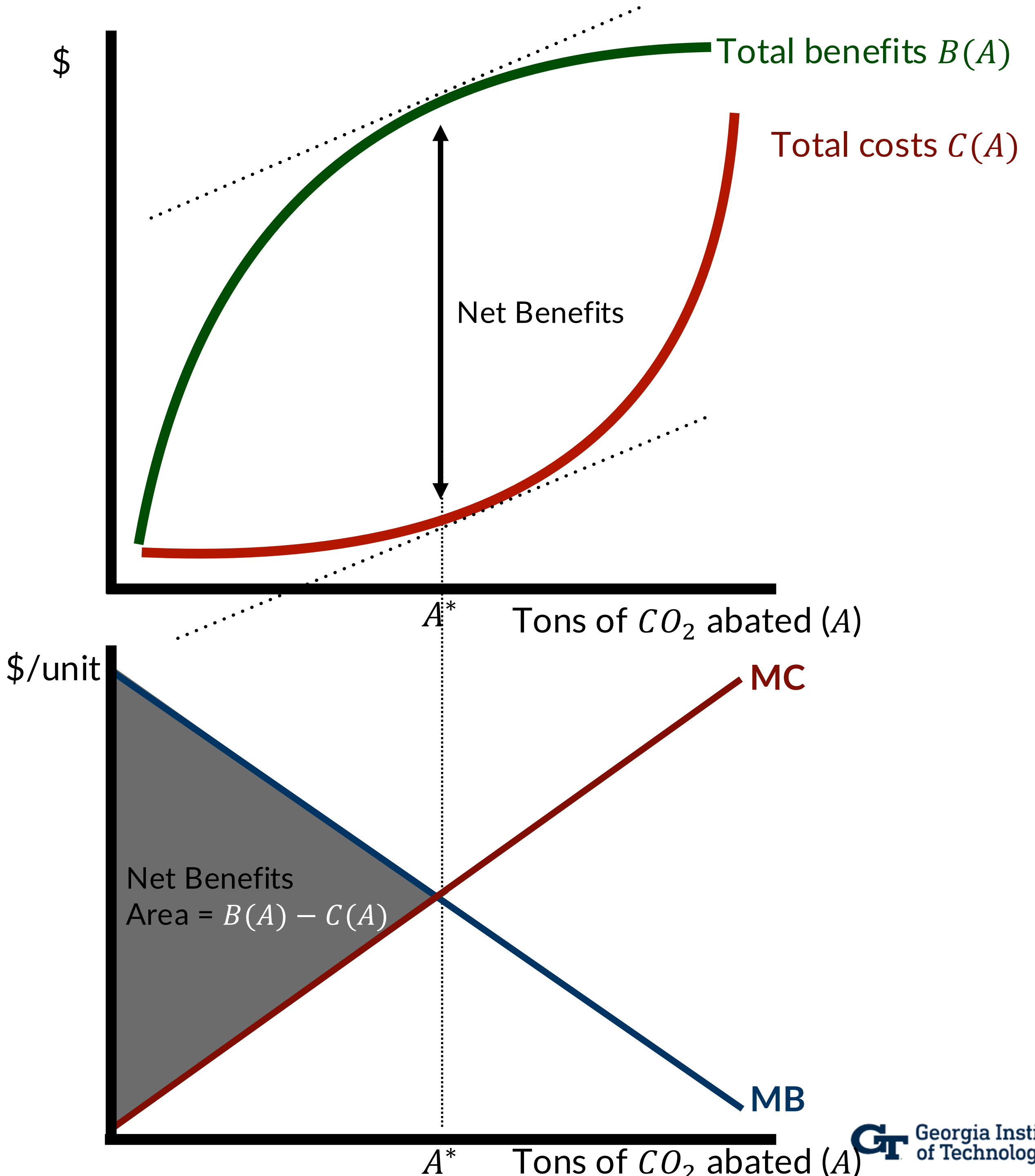
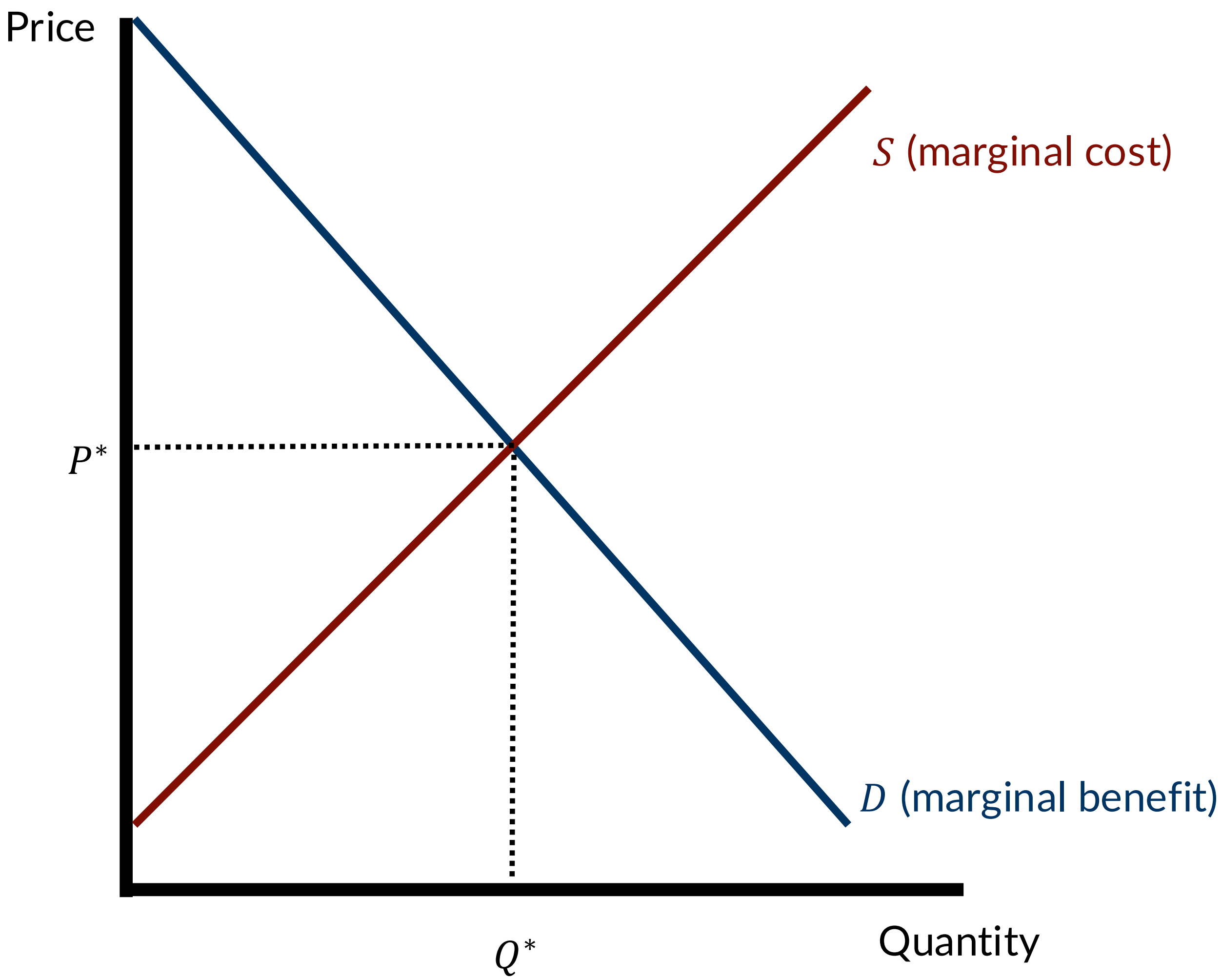
From net benefits to supply and demand

- MB is decreasing because $B(A)$ is increasing at a decreasing rate
  - This is referred to as "diminishing marginal utility"
- MC is increasing because $C(A)$ is increasing at an *increasing* rate.
  - This is referred to as "diminishing marginal returns"
- If we move to the left of A*, total benefits shrink, because MB>MC.
  - We value an additional unit of abatement more than it costs!
- If we move to the right of A*, total benefits shrink, because MC>MB.
  - It cost more to abate the last unit than its marginal value.

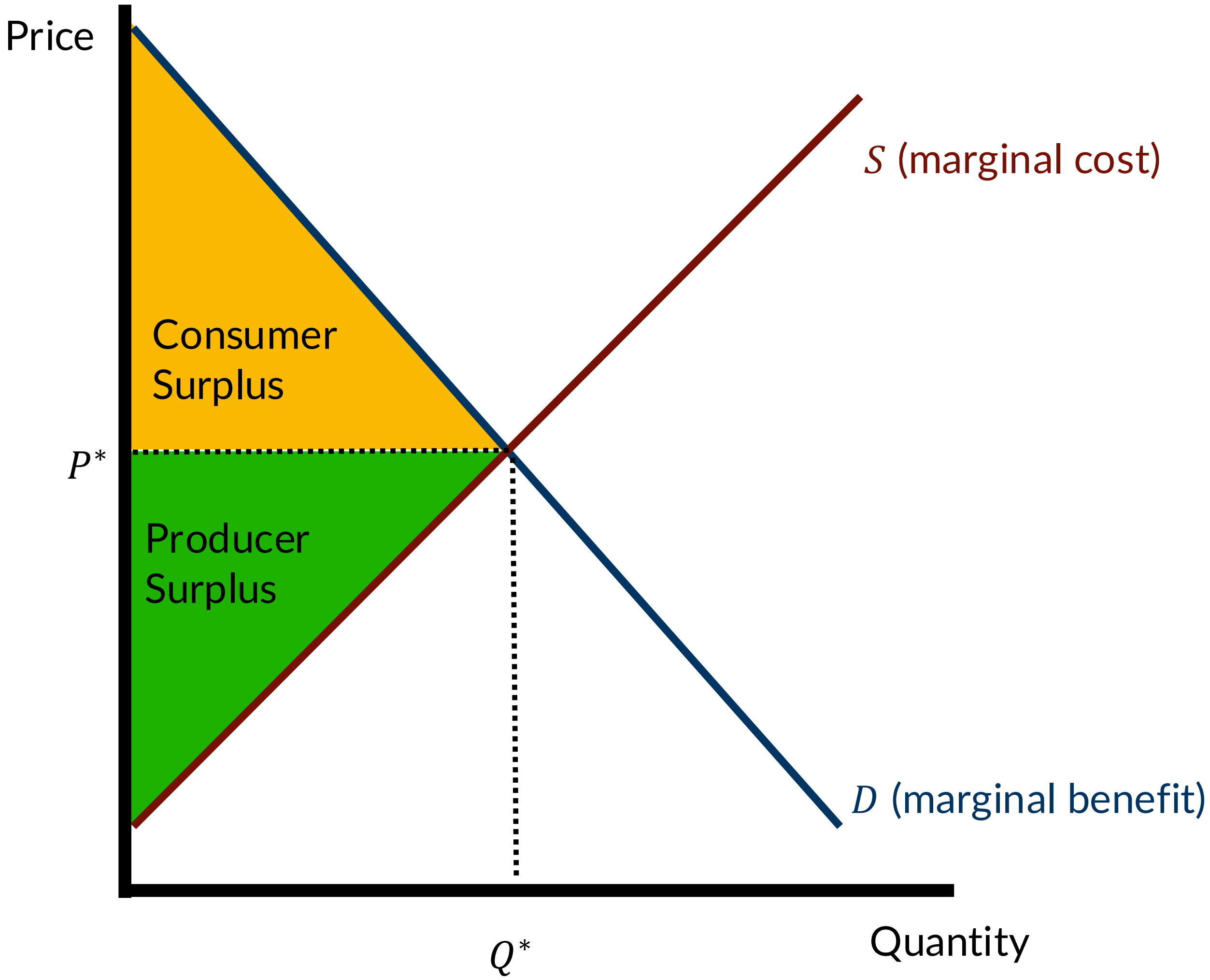


Getting back to supply and demand…



- Our demand curve traces out our willingness to pay for a particular good.
  - In other words, how much of that good would we consume at a given price?
  - Demand represents the marginal benefits of consuming one additional unit.
  - **Demand=MB.**
- Similarly, our supply curve traces out how much producers want to sell at a given price.
  - Suppliers won't sell an additional good for less than it costs to make it, so…
  - **Supply=MC.**

…and, we're back to where we started!



- In a **competitive market**, we end up at an equilibrium point where buyers and sellers "magically" end up maximizing net benefits.
  - Some of these benefits go to buyers, and some go to sellers.
- Consumer surplus represents the difference between what you would have paid relative to what you actually paid.
- Producer surplus represents the difference between what you were paid for the good less the minimum amount you would have accepted.
- The "invisible hand" allows us to maximize net benefits in a decentralized way.

When are markets efficient?

From Keohane & Olmstead…

A market equilibrium is efficient if…

1. The market is *competitive* (i.e., buyers and sellers take prices as given)
2. Firms and consumers both have *perfect information* about product quality
3. The market is *complete* (i.e., all relevant costs and benefits are borne by participants in the market).

- The 3rd component (complete markets) requires that **No Externalities are Allowed**

Climate change is an economic problem because it is the result of GHG emissions…

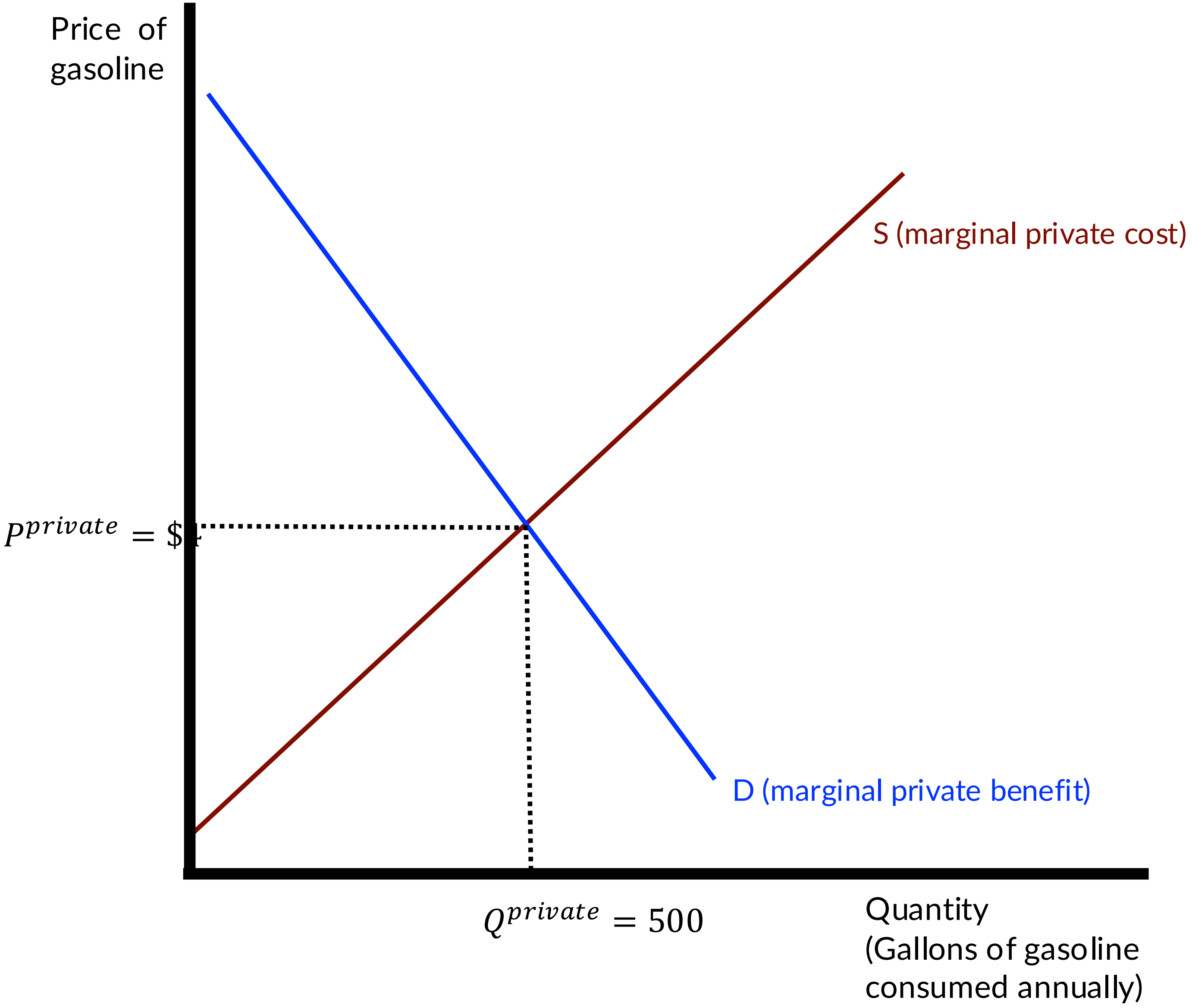
…and GHG emissions are a giant *externality.*

Externality.

An externality results when the actions of one individual (or firm) have a direct, unintentional, and uncompensated effect on the well-being of other individuals or the profits of other firms.




GHG emissions as an externality



- We typically think of climate change as the result of a negative externality from overconsumption of CO2-intensive goods and services.

- Let's consider the retail market for gasoline.
- At $3/gal, I am willing to purchase about 500 gallons of gas a year.

- We know gasoline emits CO2 (among other things) when we burn it. But, I need to get to campus to teach ECON-4210, so… I am going to drive.

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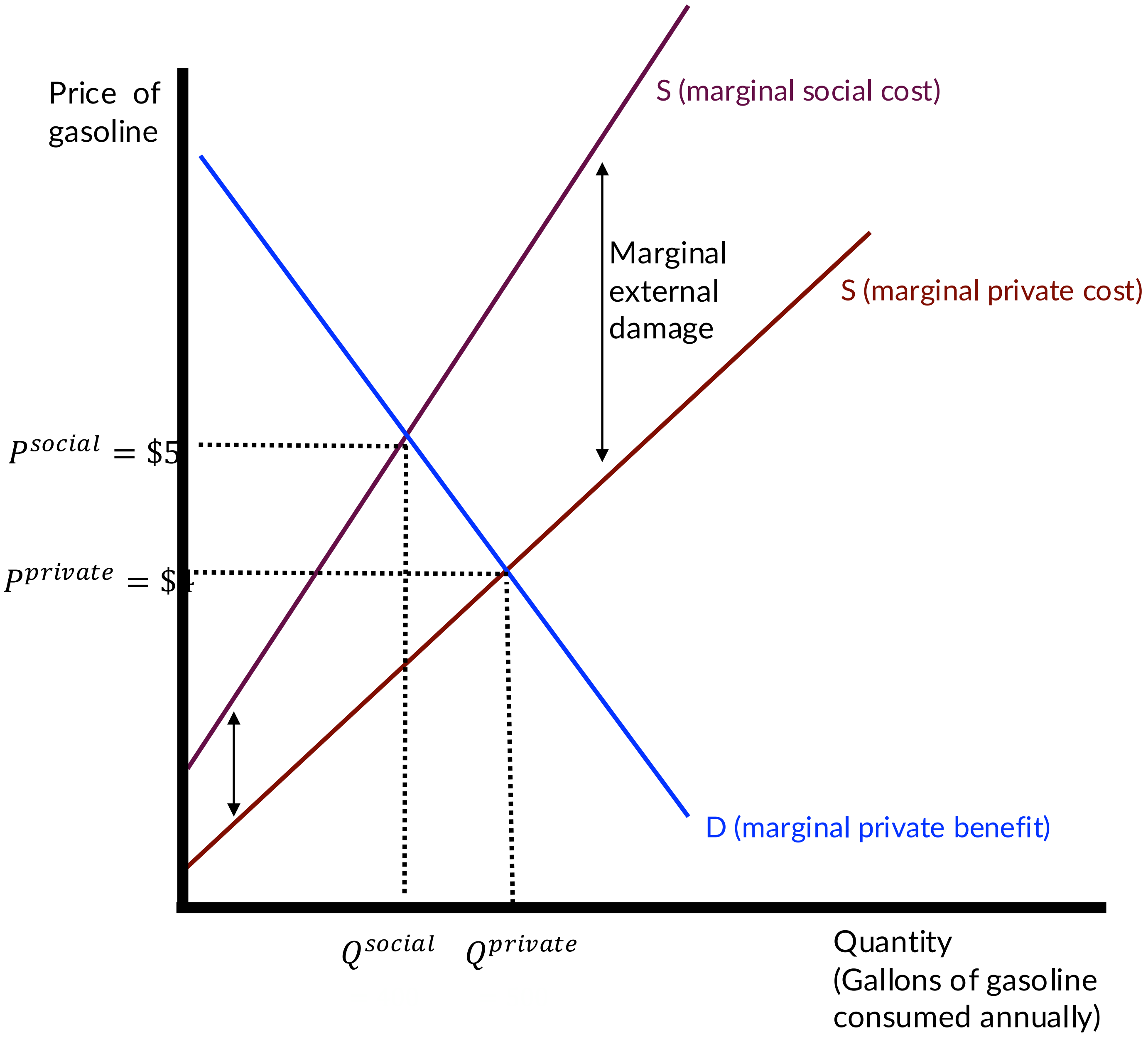
GHG emissions as an externality



- The **marginal social cost** curve captures the total cost (inclusive of any harm caused by GHG emissions of my gasoline consumption).
- This additional cost is called the **marginal external damage**. This is the vertical distance between the **private MC** and the **social MC** curves.
- I am consuming too much gasoline because I don't pay its full cost.
- The ***efficient*** amount of gasoline, that takes into account its external costs, is represented by Q^{social} at a price of $4.
- Negative externalities shift the supply curve to the NW because they are unaccounted-for costs. (Positive externalities shift demand curves to the SE because they are unaccounted for benefits).

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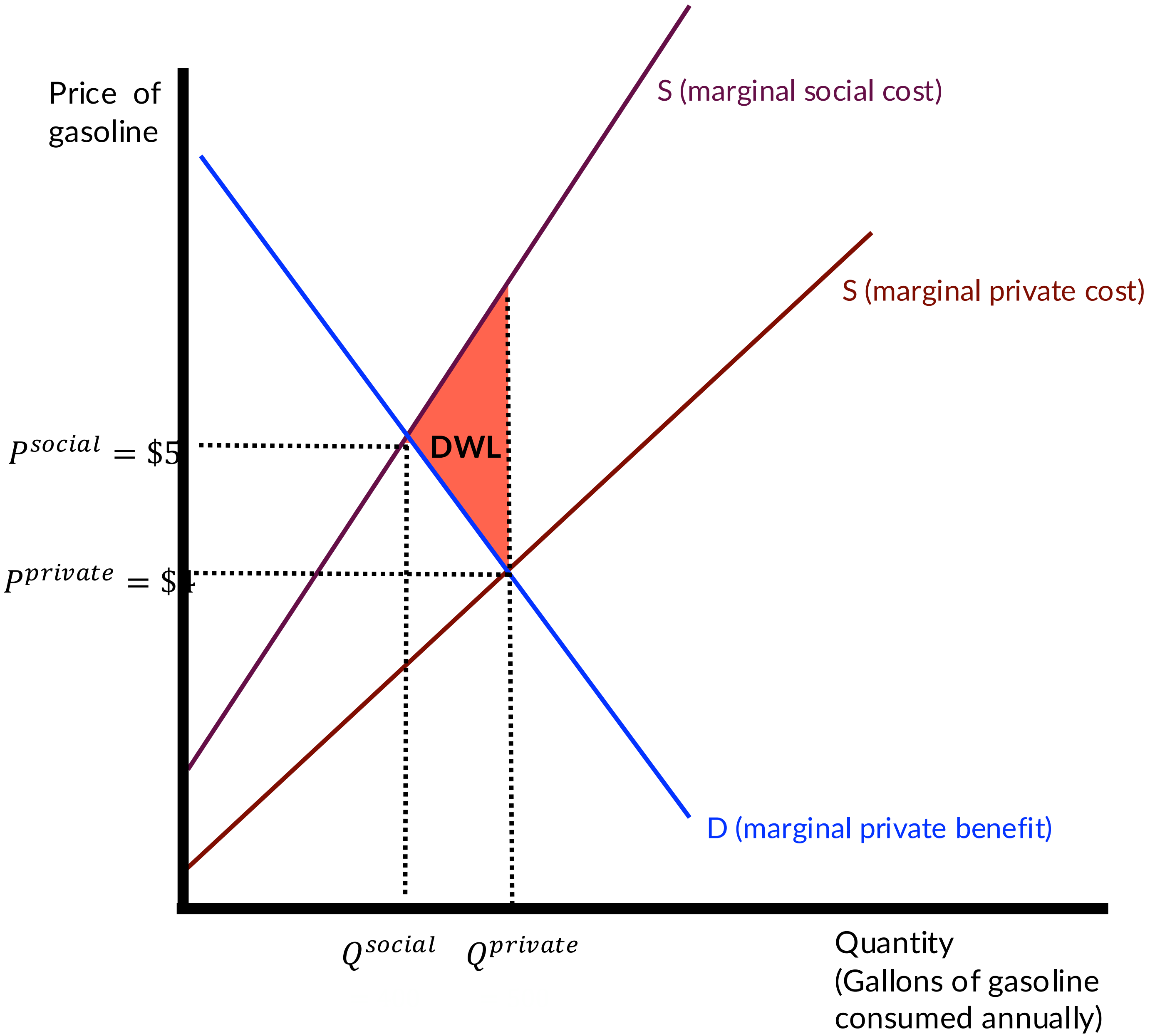
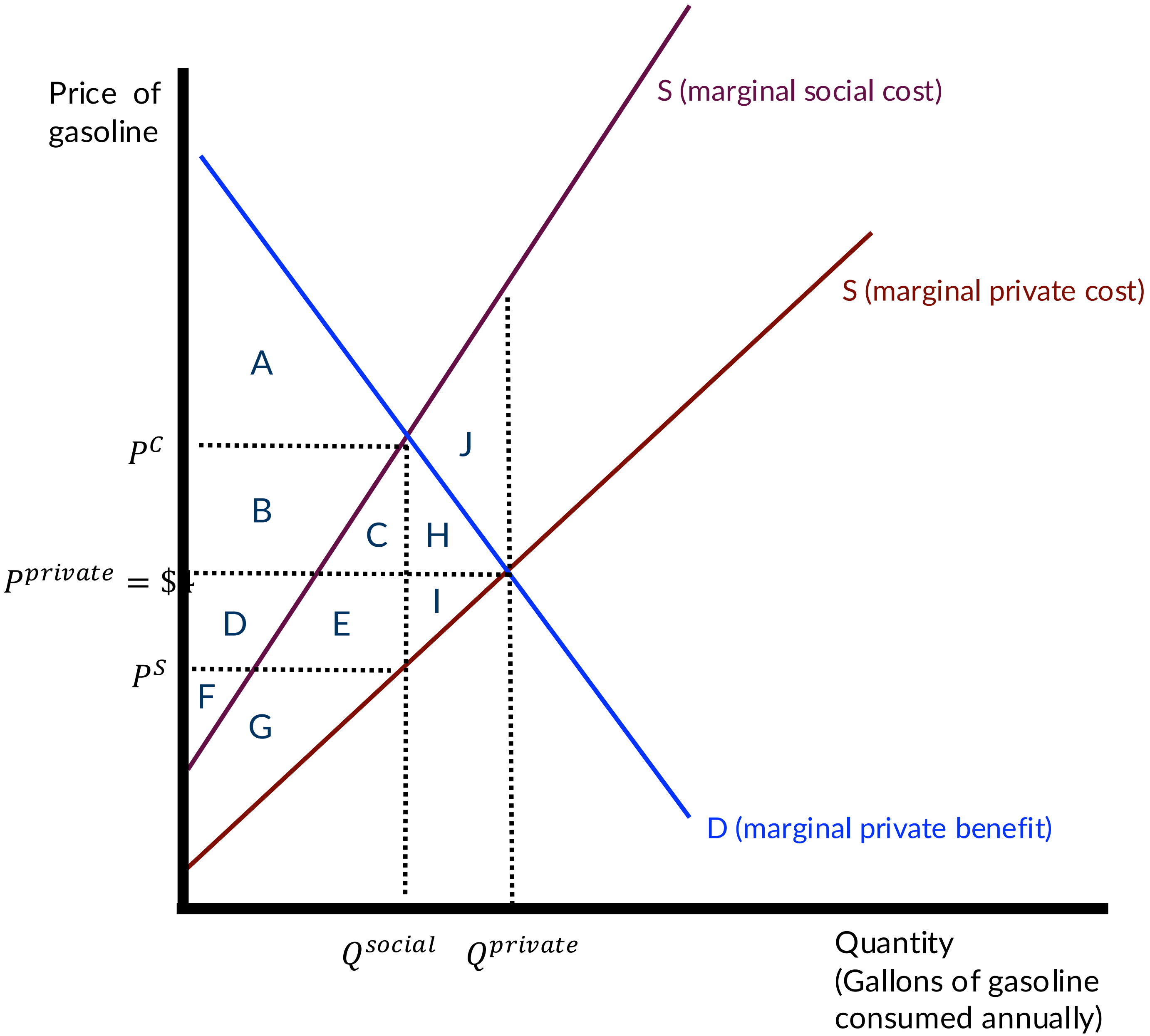
GHG emissions as an externality



- In the <u>private equilibrium</u>, I'm using too much gas and this impose consequences on society. How big are those consequences?
- **Deadweight loss** captures the welfare cost of consuming too many GHGs.
- When we account for marginal social cost, we incorporate the welfare of others who are affected by my consumption of gasoline. This is what DWL represents.
- When we move from $Q^{private}$ to Q^{social} we "lose" DWL (which is a good thing!)
- Overall welfare will increase because we'll eliminate **DWL**.

GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus			
Consumer Surplus			
Externality Cost			
Gov't Revenue			
Total Surplus			

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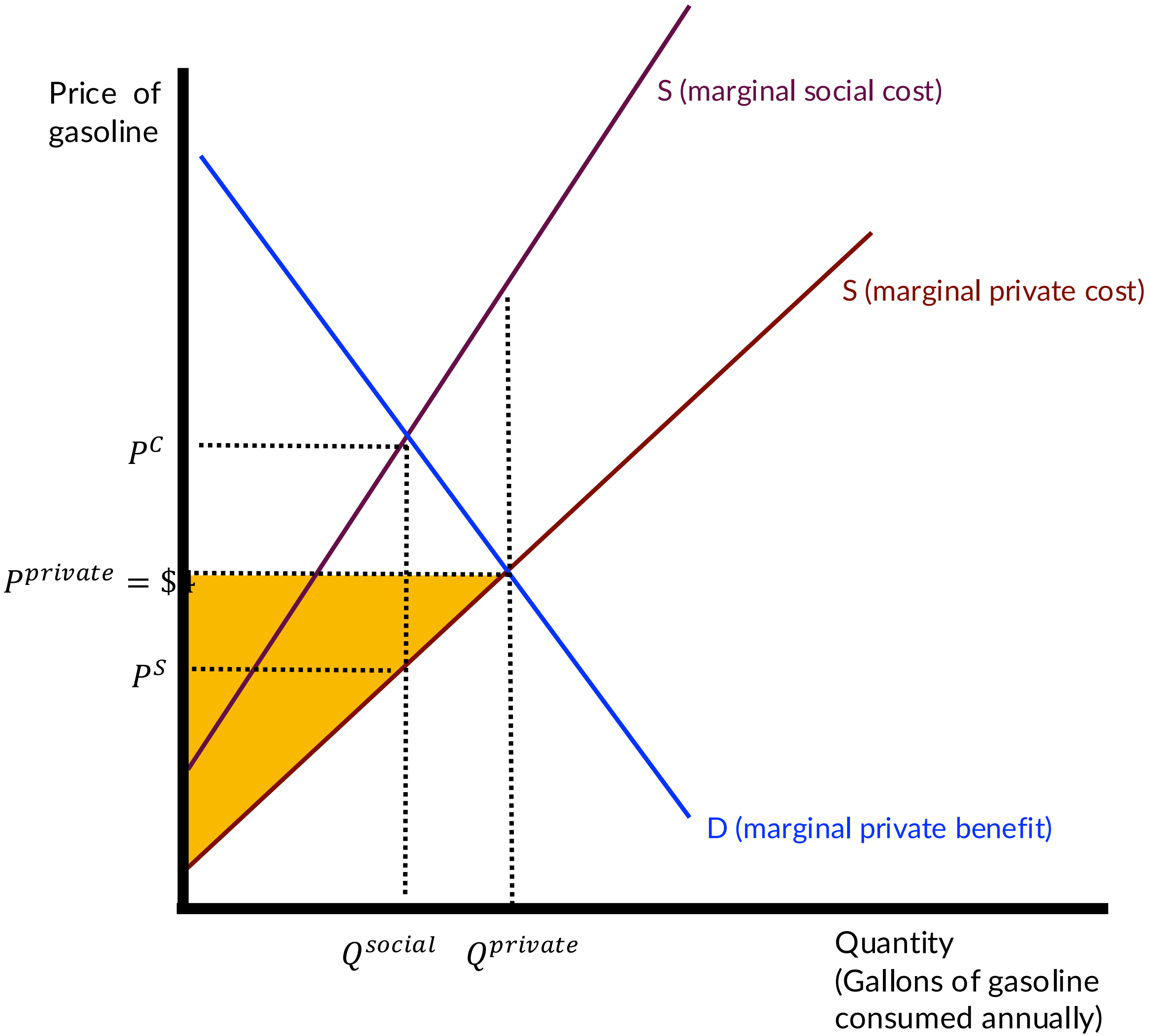
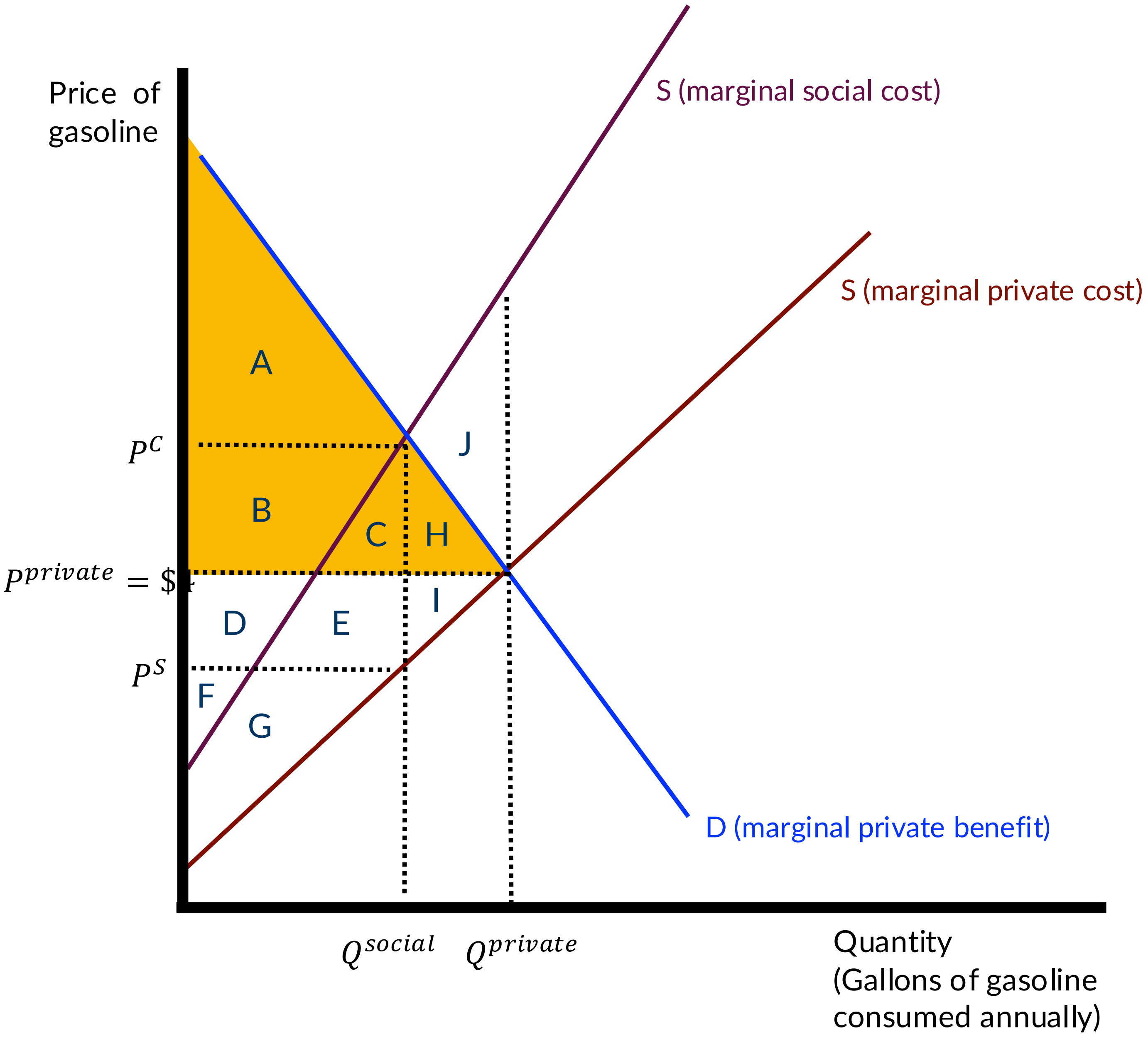
GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G		
Consumer Surplus			
Externality Cost			
Gov't Revenue			

GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G		
Consumer Surplus	A+B+C+H		
Externality Cost			
Gov't Revenue			
Total Surplus			

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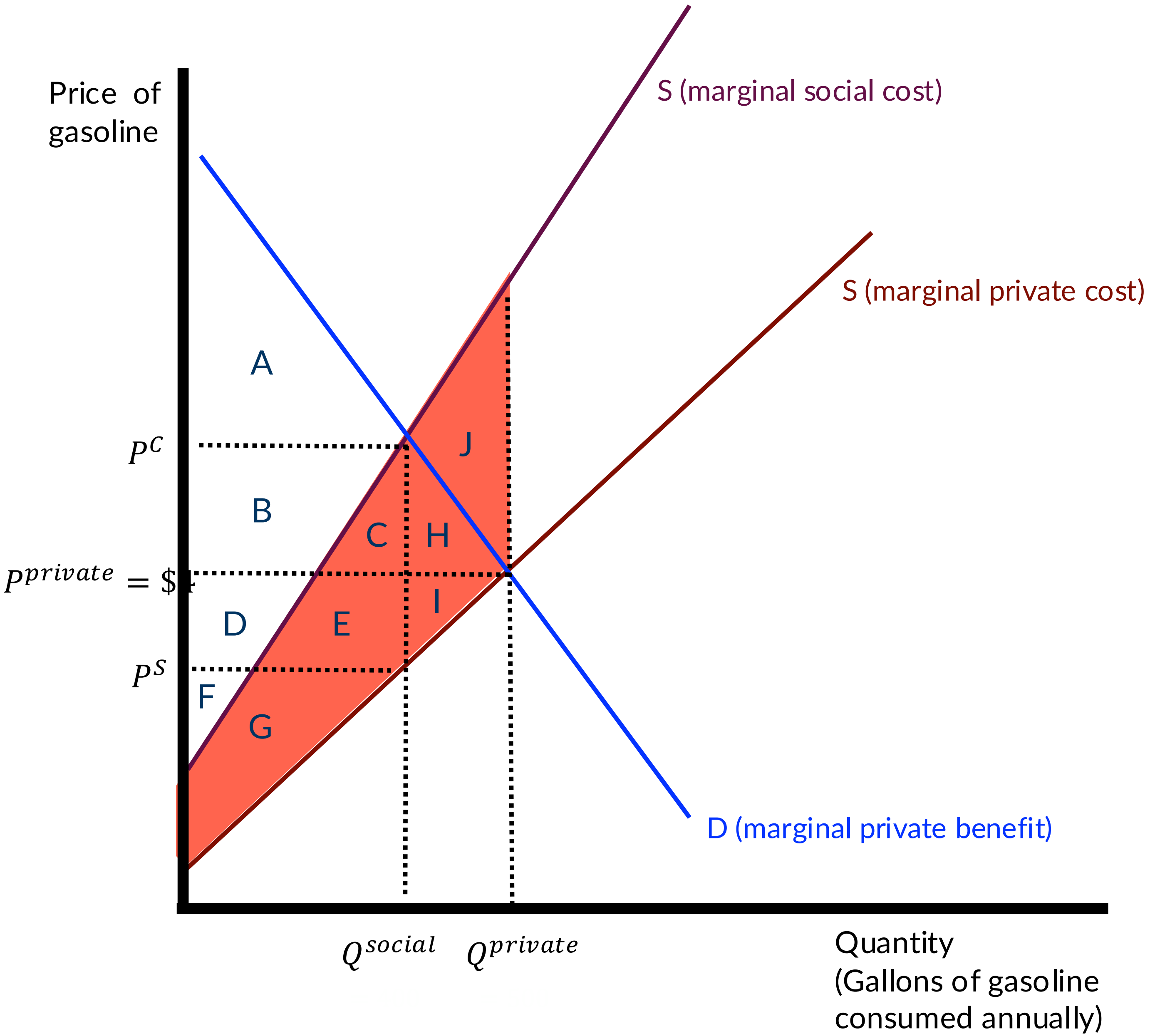
GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G		
Consumer Surplus	A+B+C+H		
Externality Cost	−(G+E+C +H+I+J)		
Gov't Revenue			
Total Surplus			

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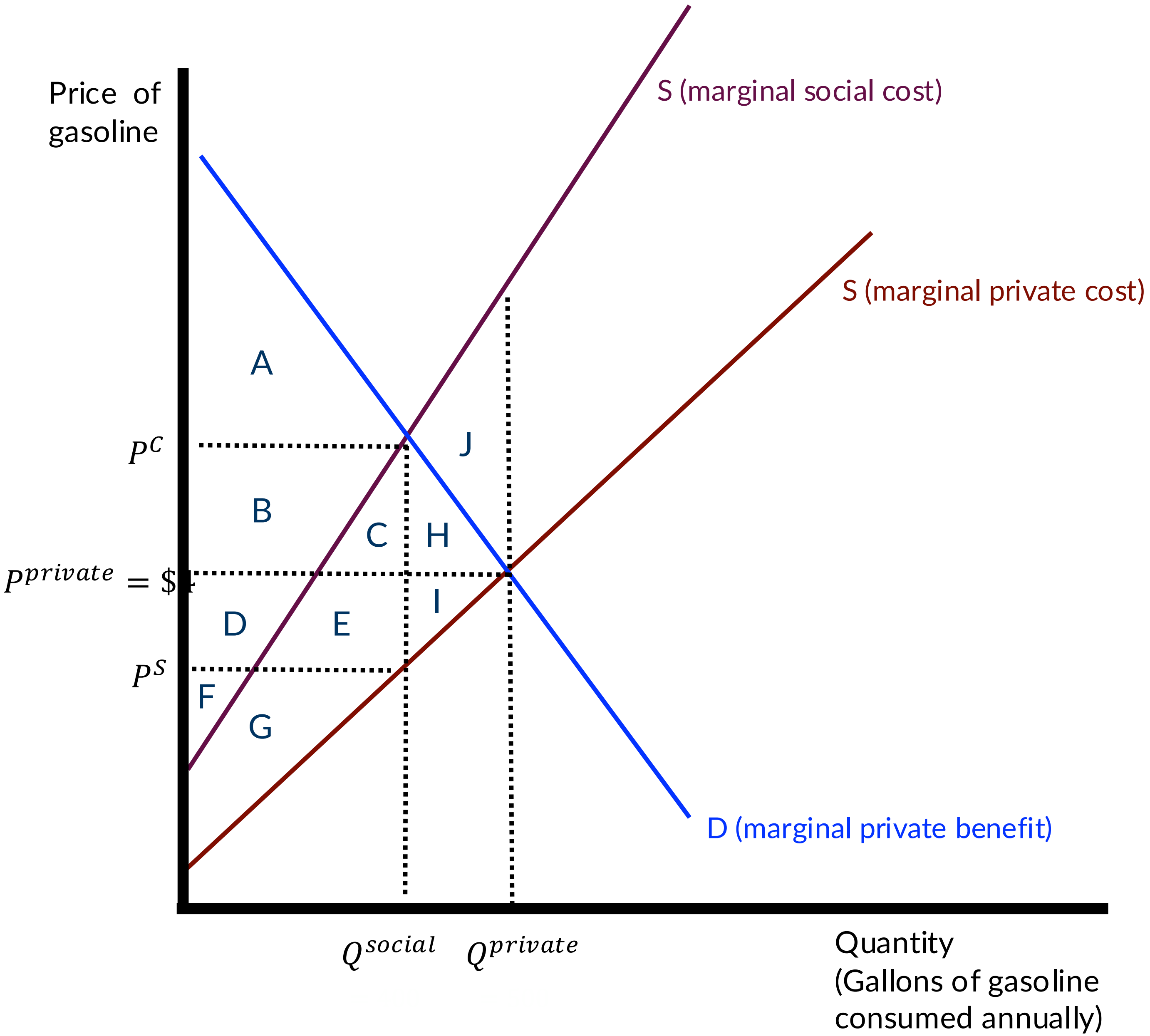
GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G		
Consumer Surplus	A+B+C+H		
Externality Cost	−(G+E+C +H+I+J)		
Gov't Revenue	0		
Total Surplus			

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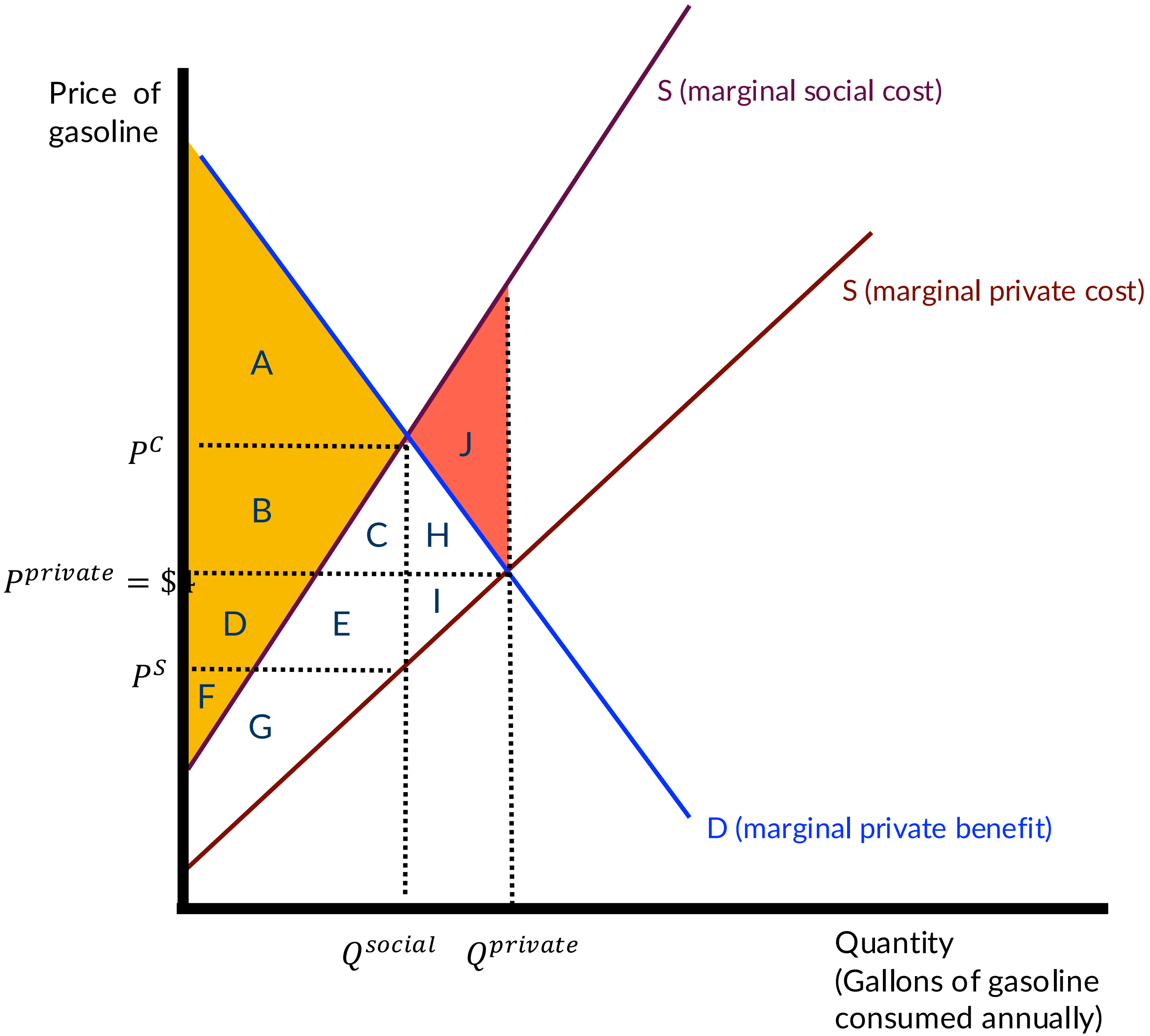
GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G		
Consumer Surplus	A+B+C+H		
Externality Cost	−(G+E+C +H+I+J)		
Gov't Revenue	0		
Total Surplus	D+F+A+B−J		

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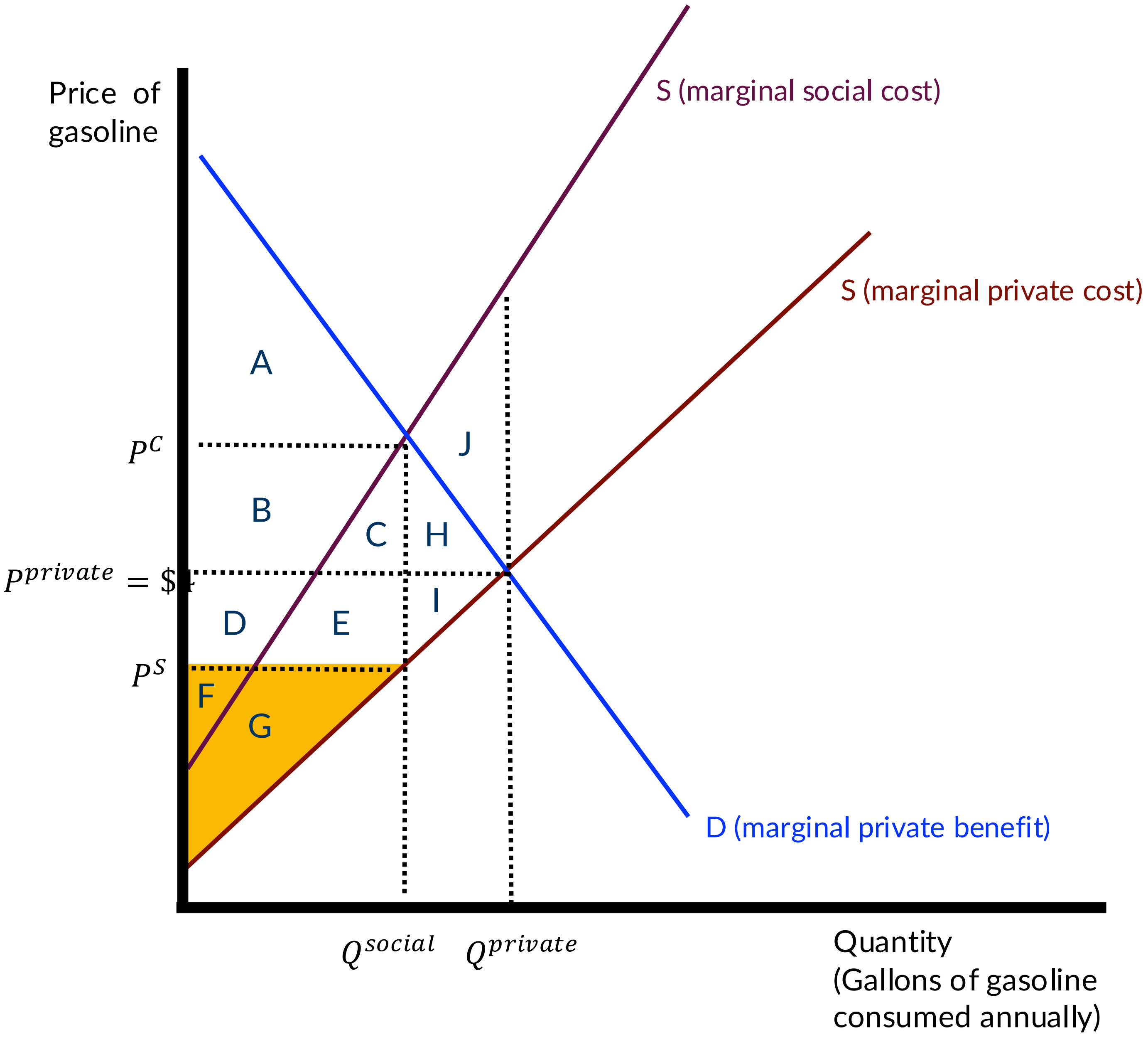
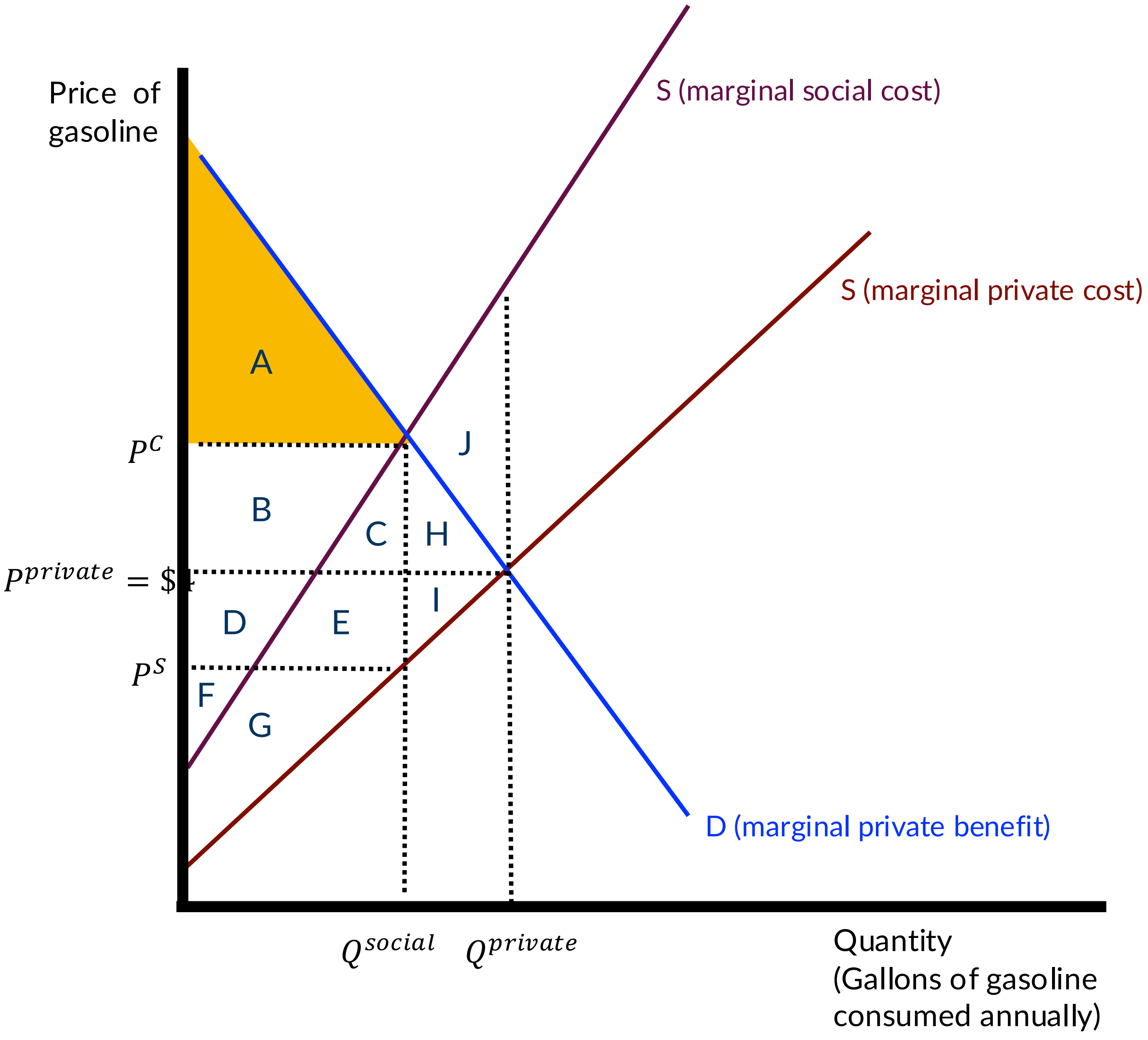
GHG emissions as an externality



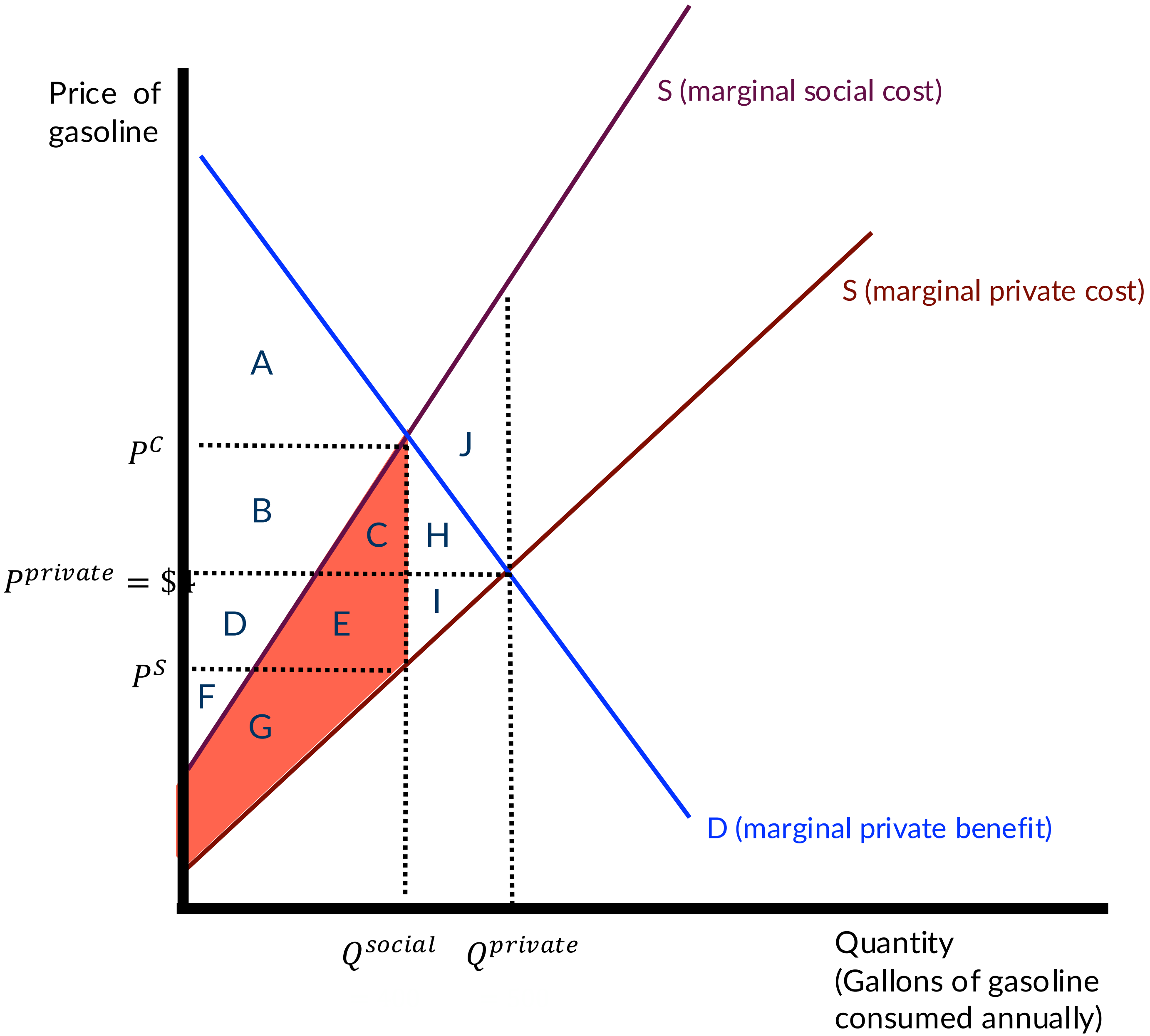
	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G	F+G	
Consumer Surplus	A+B+C+H		
Externality Cost	–(G+E+C +H+I+J)		
Gov't Revenue	0		
Total Surplus	D+F+A+B–J		

GHG emissions as an externality



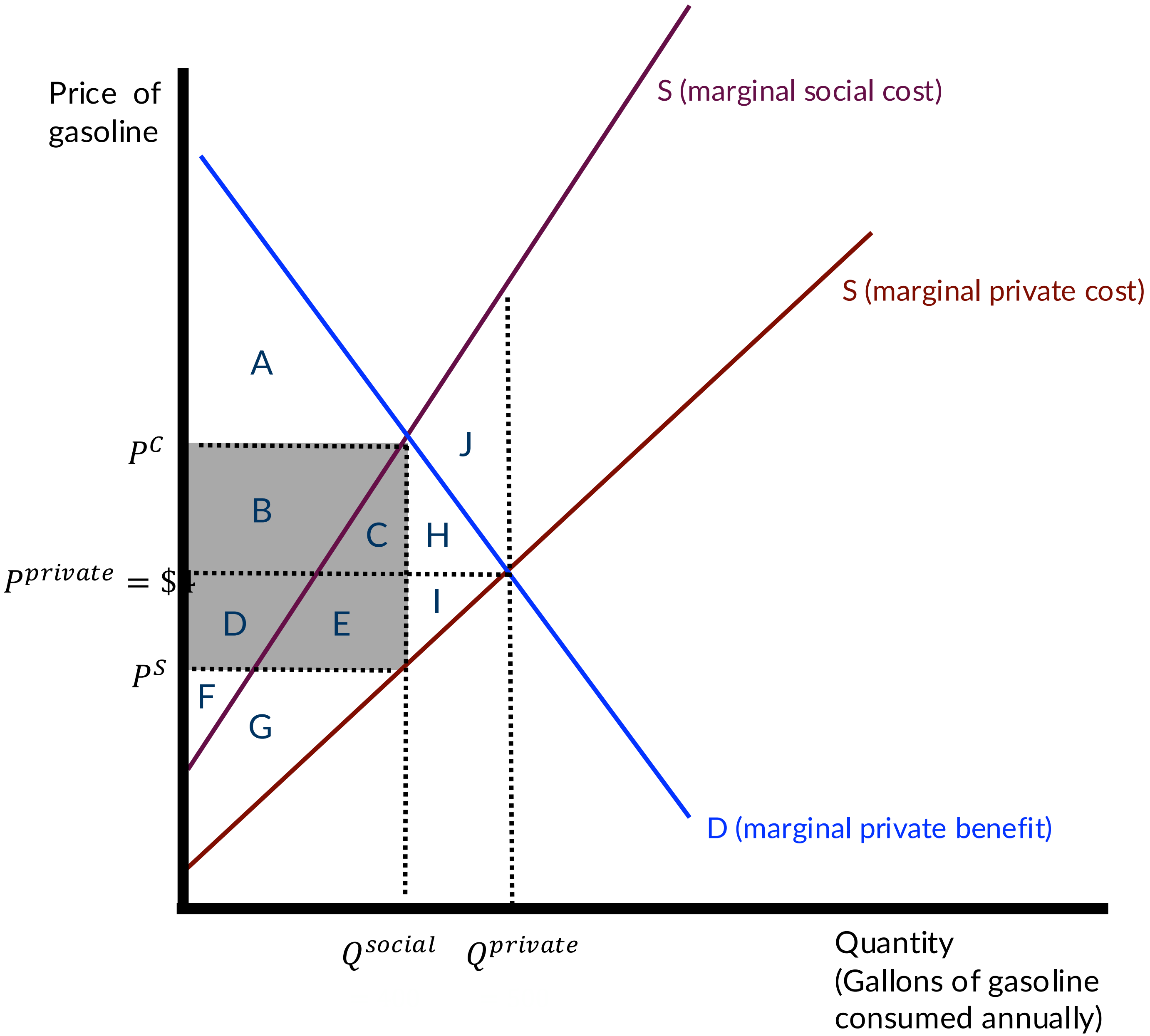
	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G	F+G	
Consumer Surplus	A+B+C+H	A	
Externality Cost	–(G+E+C +H+I+J)		
Gov't Revenue	0		
Total Surplus	D+F+A+B–J		

GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G	F+G	
Consumer Surplus	A+B+C+H	A	
Externality Cost	−(G+E+C +H+I+J)	−(G+E+C)	
Gov't Revenue	0		
Total Surplus	D+F+A+B−J		

GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G	F+G	
Consumer Surplus	A+B+C+H	A	
Externality Cost	−(G+E+C +H+I+J)	−(G+E+C)	
Gov't Revenue	0	B+C+D+E	
Total Surplus	D+F+A+B−J		

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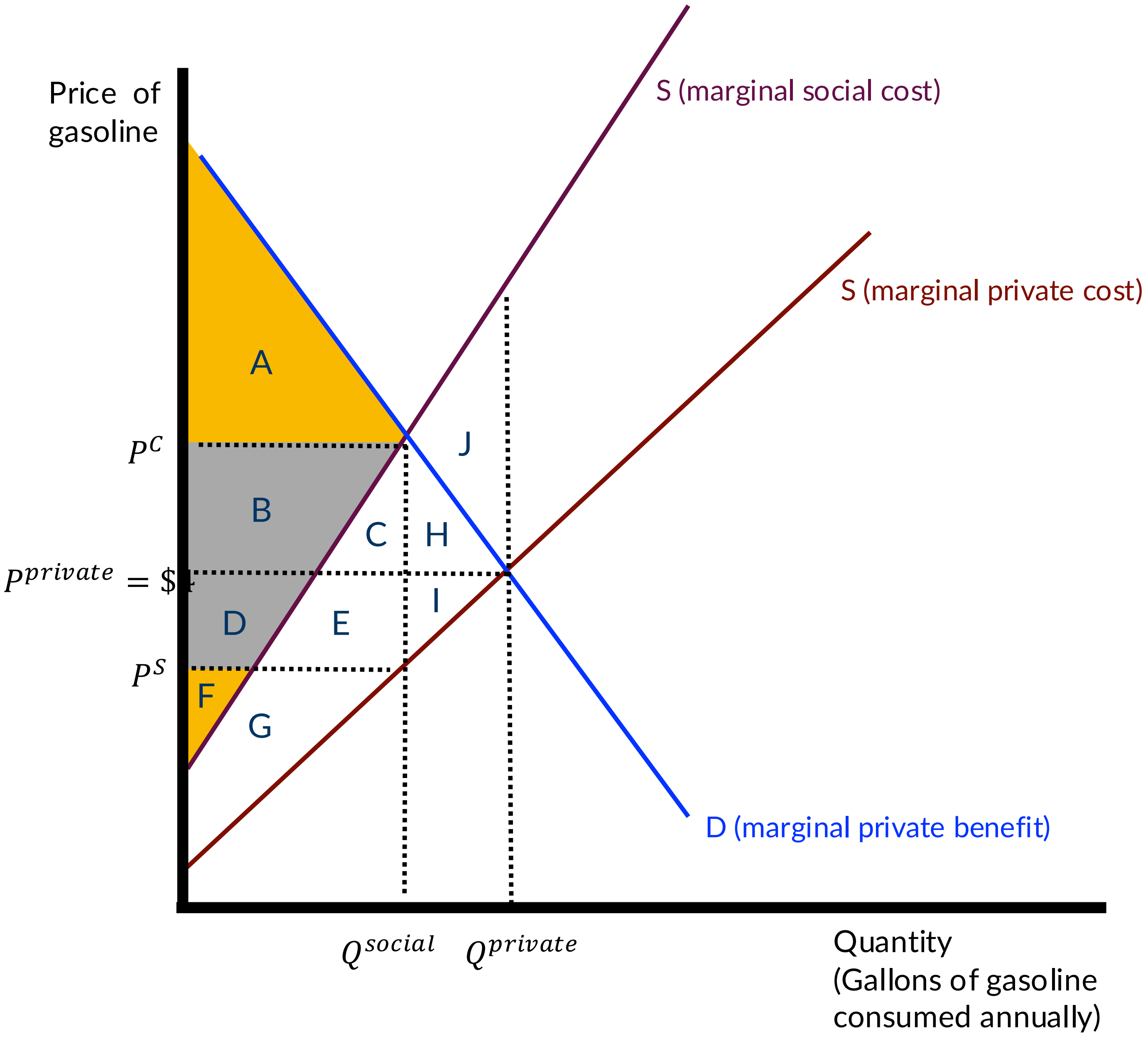
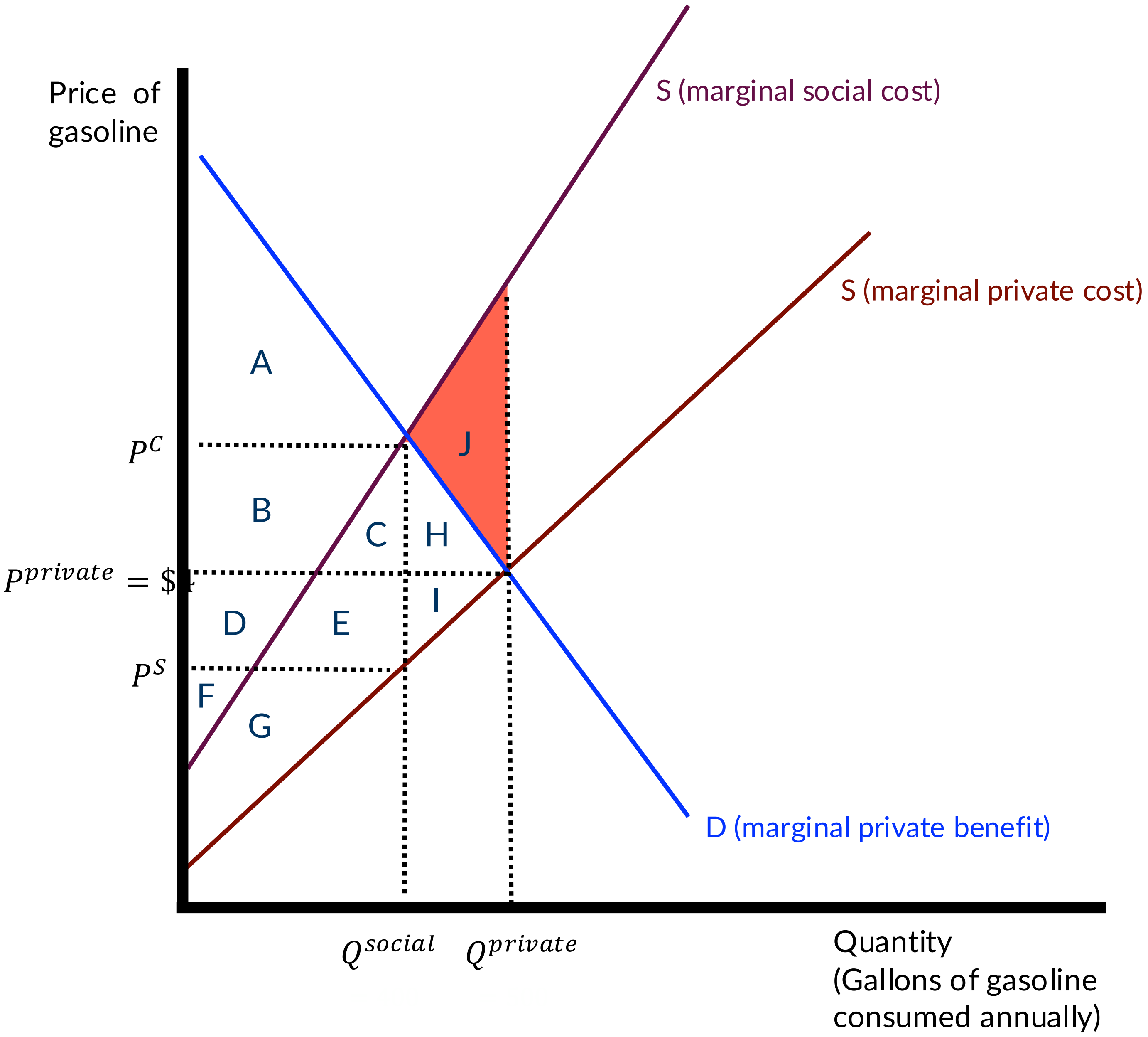
GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G	F+G	
Consumer Surplus	A+B+C+H	A	
Externality Cost	–(G+E+C +H+I+J)	–(G+E+C)	
Gov't Revenue	0	B+C+D+E	
Total Surplus	D+F+A+B–J	F+A+B+D	

GHG emissions as an externality



	Competitive (private) equilibrium	Socially optimal equilibrium	Diff.
Producer Surplus	D+E+I+F+G	F+G	D+E+I
Consumer Surplus	A+B+C+H	A	B+C+H
Externality Cost	−(G+E+C +H+I+J)	−(G+E+C)	−(H+I+J)
Gov't Revenue	0	B+C+D+E	−(B+C +D+E)
Total Surplus	D+F+A+B−J	F+A+B+D	DWL = −J

For next week…

Have a great (long) weekend!

Read: Keohane & Olmstead, Chs. 2, 4 & 5 (continued)
www.caseyjwichman.com/econ4210#week3